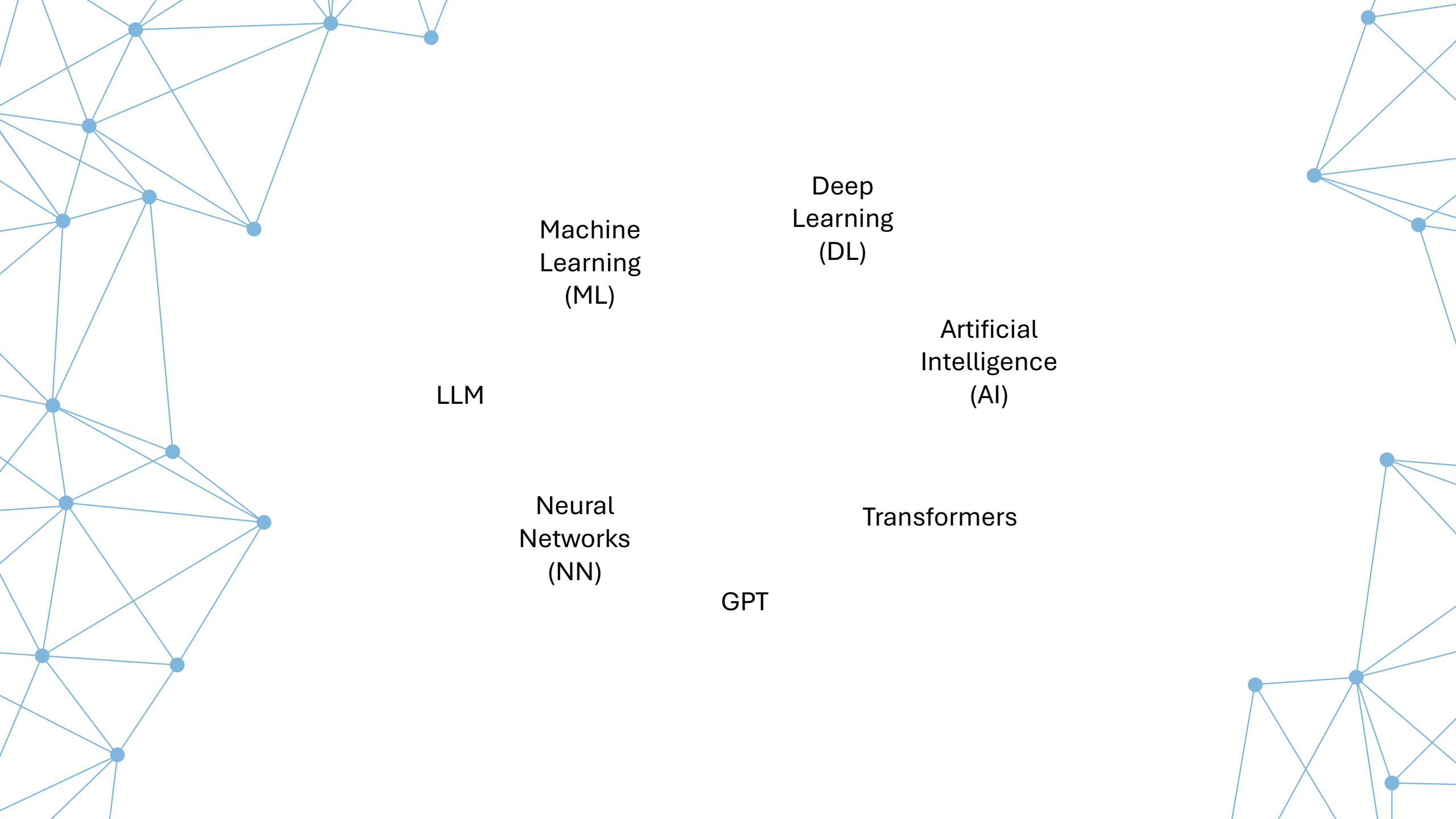




Anatomia di un Large Language Model

Dalle reti neurali all'illusione del ragionamento;
cosa c'è dietro al prompt?

Stefano Simonazzi



Machine
Learning
(ML)

Deep
Learning
(DL)

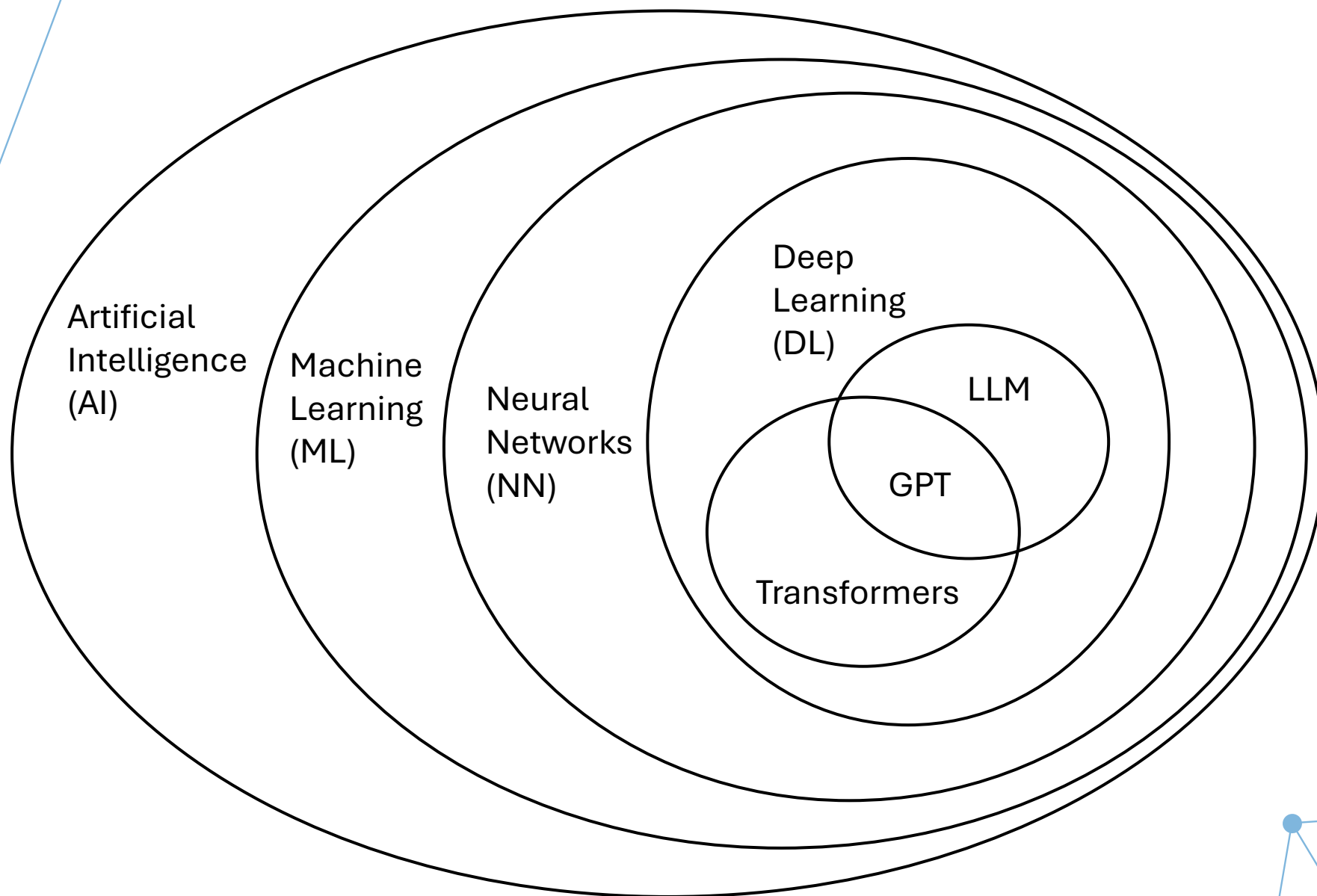
Artificial
Intelligence
(AI)

Transformers

Neural
Networks
(NN)

LLM

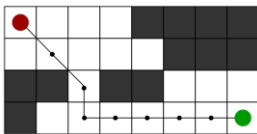
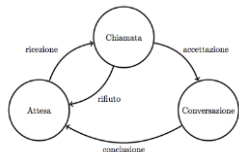
GPT



Artificial Intelligence (AI)

1956

John McCarthy
Marvin Minsky
Claude Shannon
Alan Turing (1912)
Leibniz (1646)



Sistemi esperti
Deep Blue (1997)

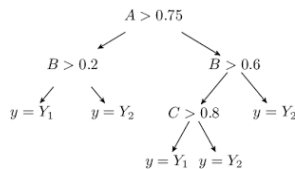


Machine Learning (ML)

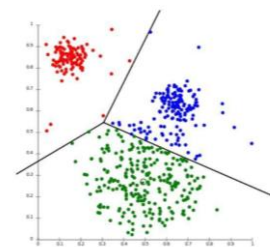
1959

Arthur Samuel

Alberi di decisione



Clustering

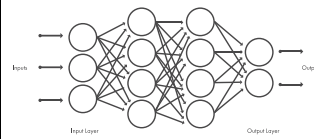


Filtri anti-spam

Neural Networks (NN)

1958

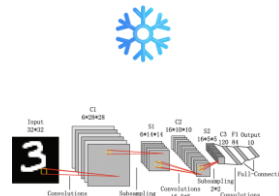
Frank Rosenblatt
John Hopfield



Deep Learning (DL)

80'

Geoffrey Hinton
Yann LeCun
Yoshua Bengio



2012

AlexNet

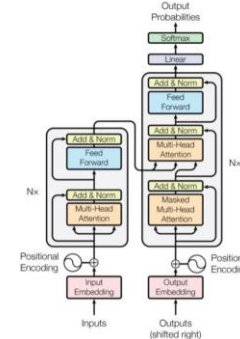
Alex Krizhevsky
Ilya Sutskever
Geoffrey Hinton



Transformers

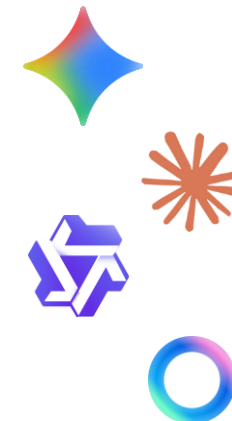
2017

Google Brain



Large Language Model (LLM)

2019

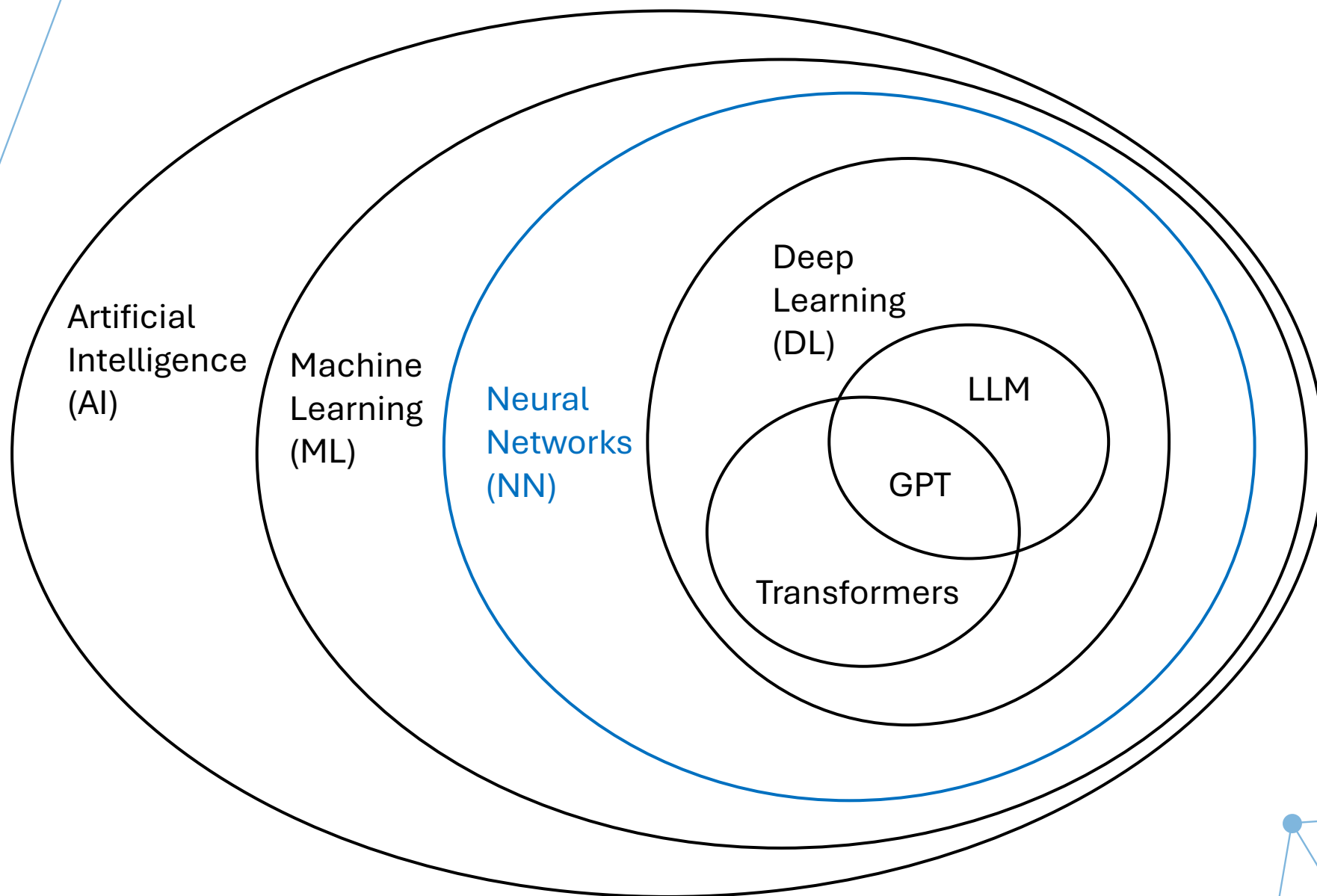


Generative Pre-trained Transformer (ChatGPT)

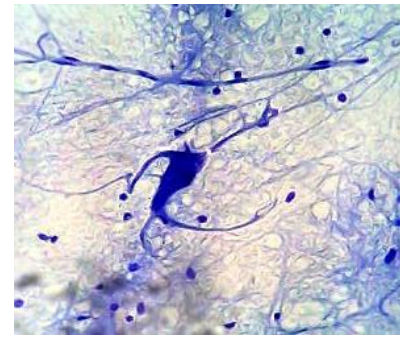
2022

OpenAI

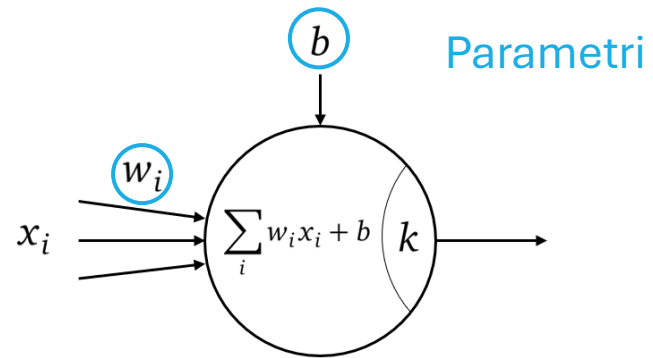




Neural Networks

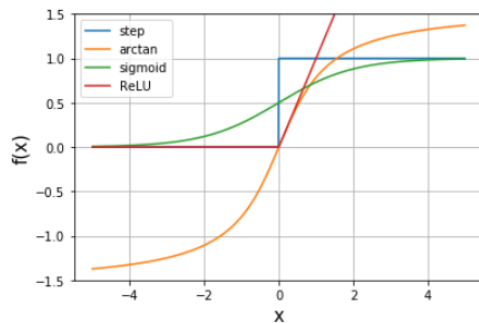


Neurone:

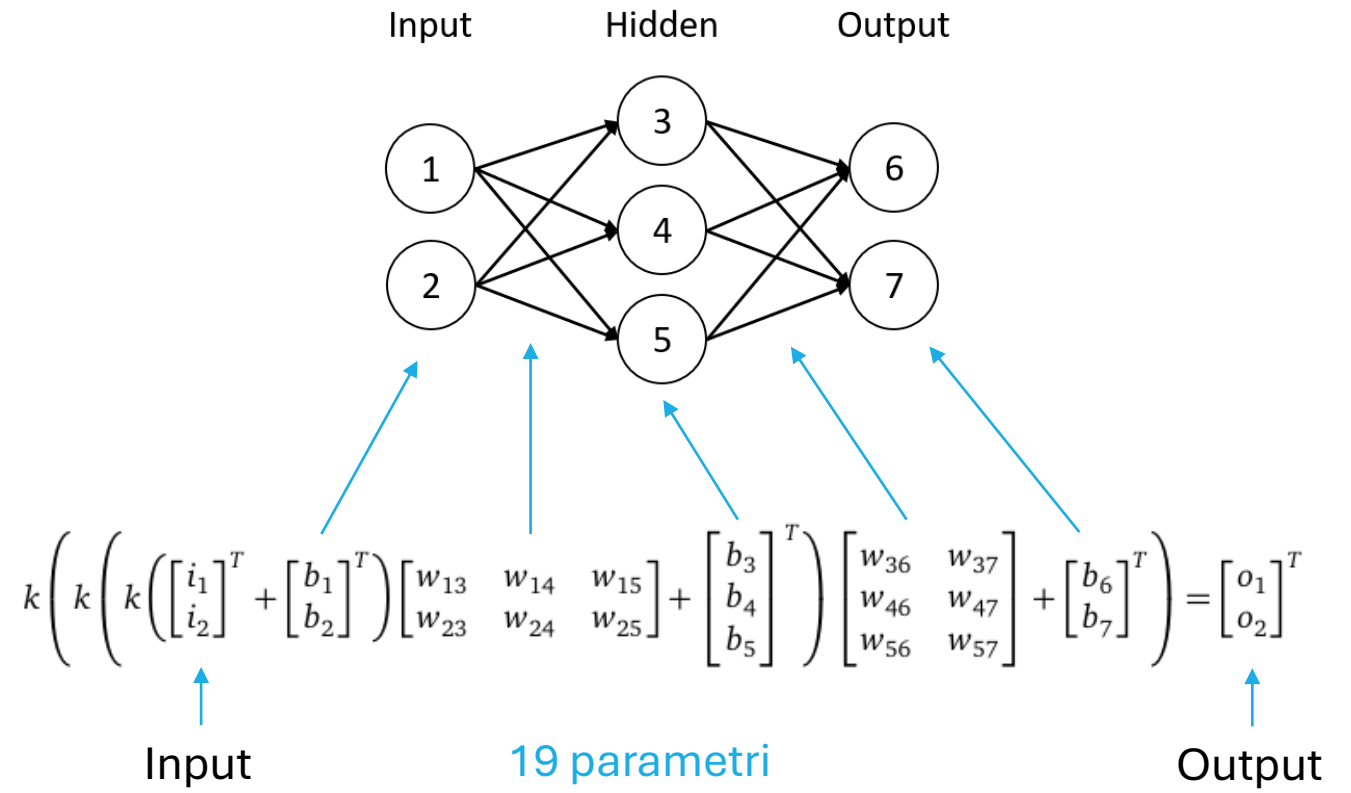


$$f(x_1, \dots, x_n) = k\left(\sum_i w_i x_i + b\right)$$

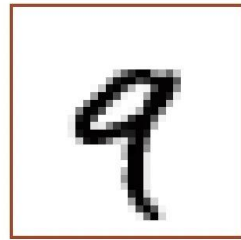
k:



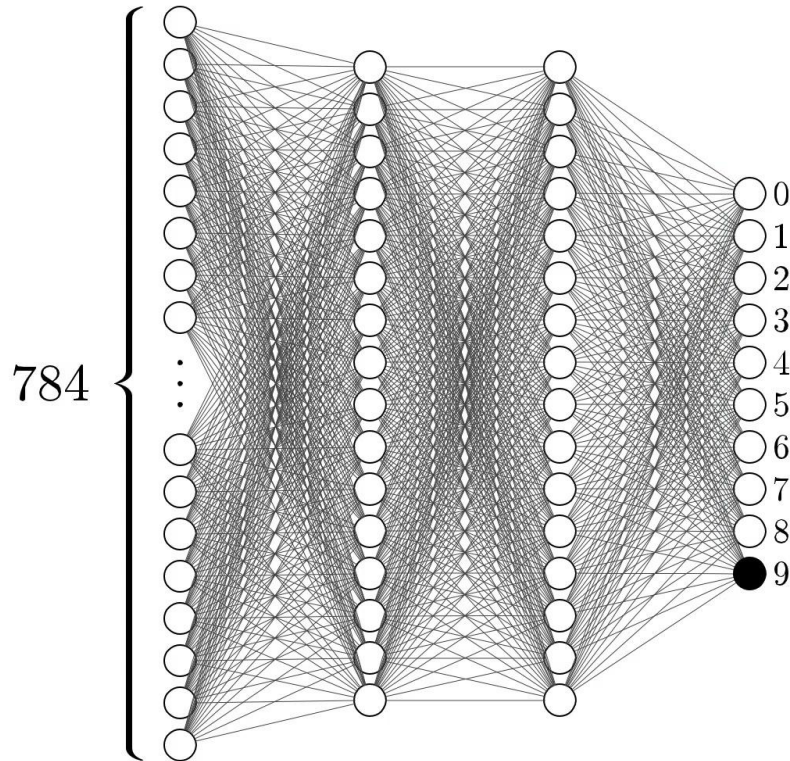
Rete neurale:



Neural Networks

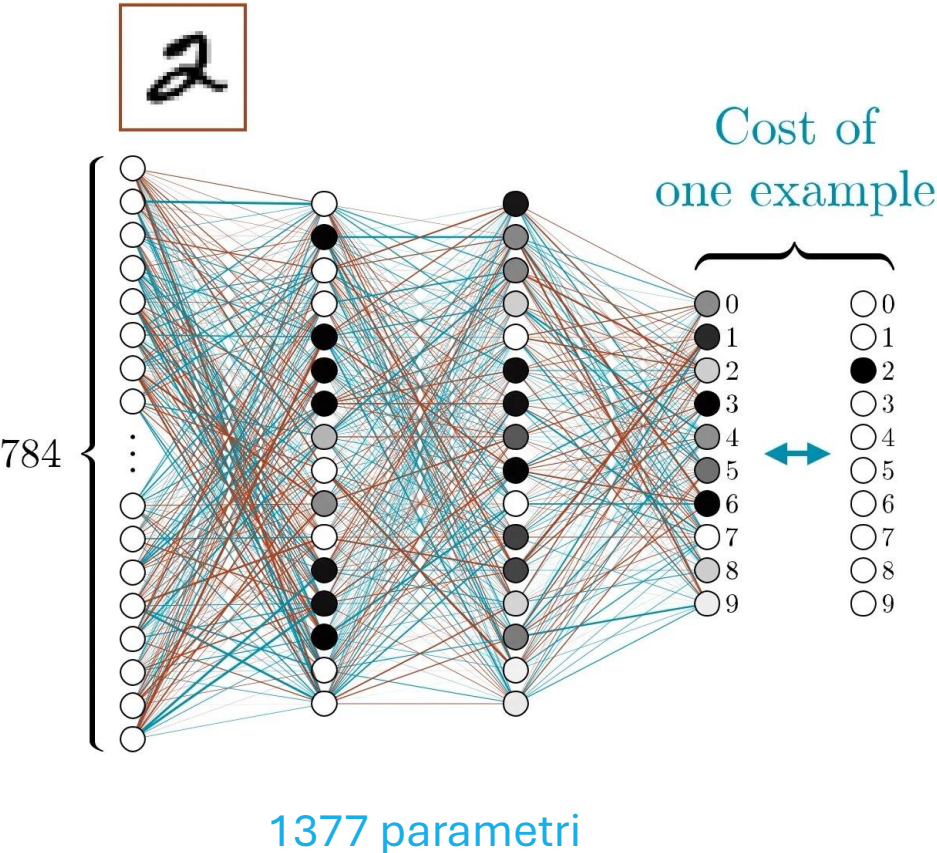


28x28 pixel

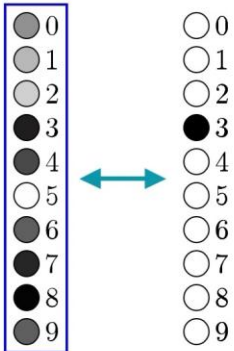


1377 parametri

Backpropagation



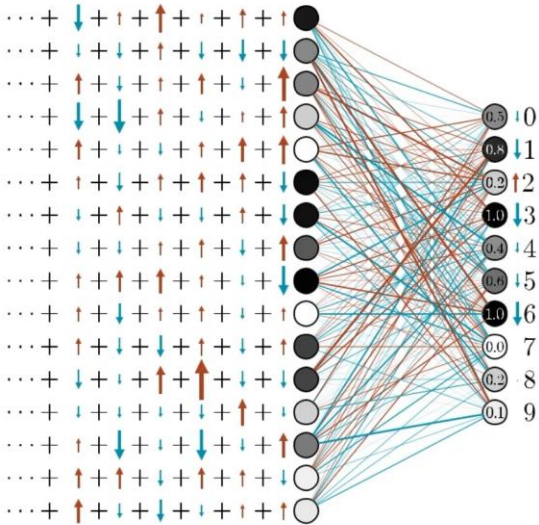
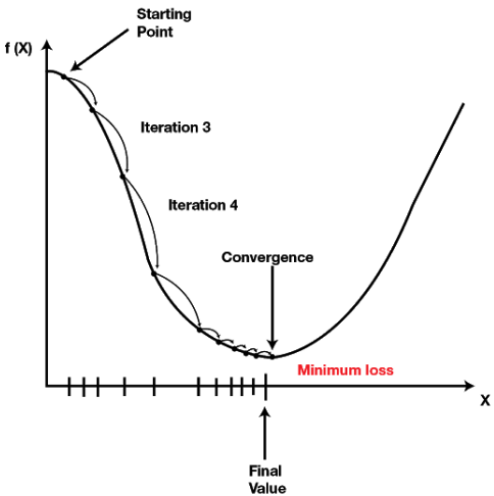
What's the "cost" of this difference?



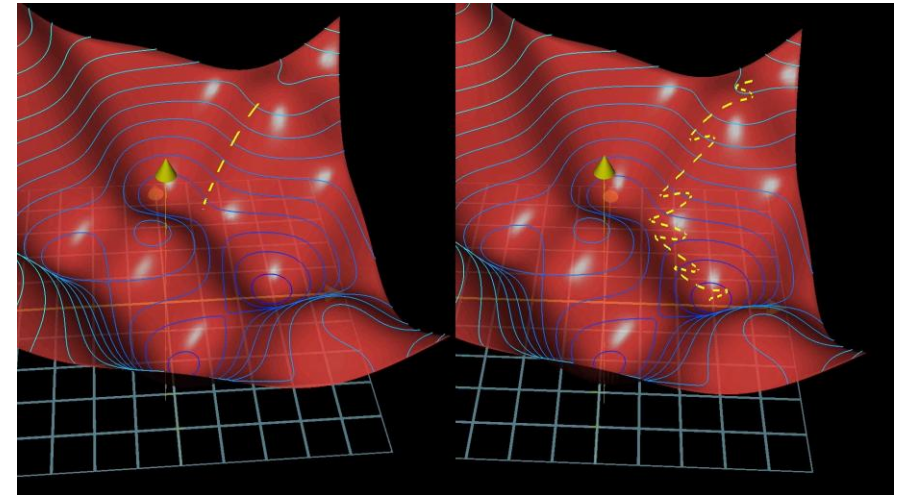
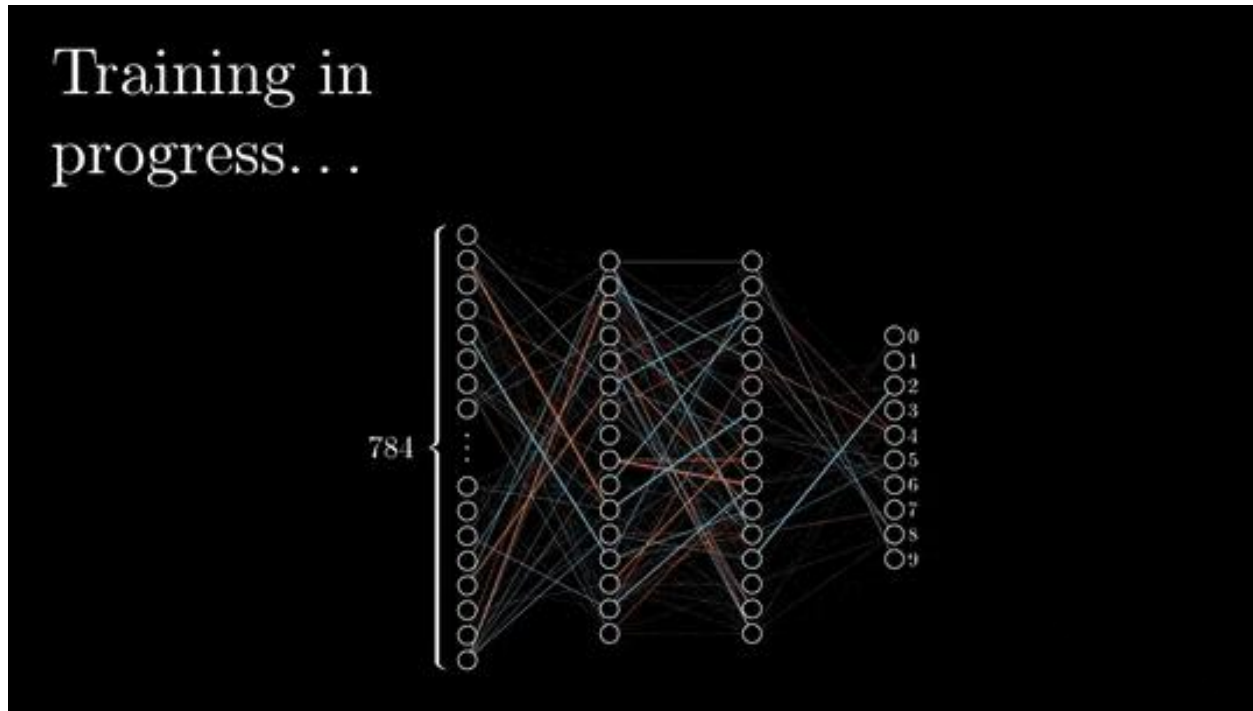
Utter trash

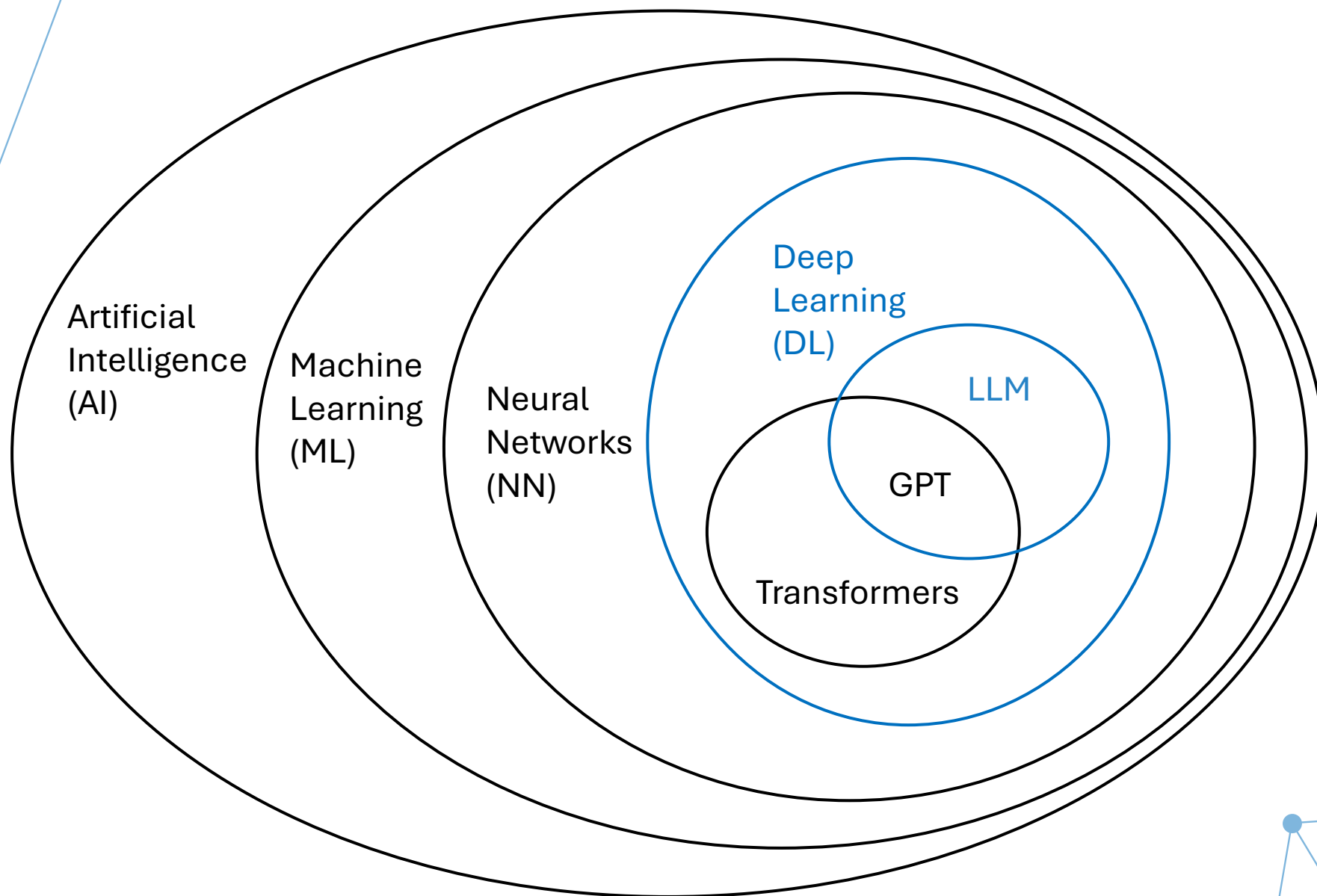
Cost of **3**

$$\left\{ \begin{array}{l} (0.43 - 0.00)^2 + \\ (0.28 - 0.00)^2 + \\ (0.19 - 0.00)^2 + \\ (0.88 - 1.00)^2 + \\ (0.72 - 0.00)^2 + \\ (0.01 - 0.00)^2 + \\ (0.64 - 0.00)^2 + \\ (0.86 - 0.00)^2 + \\ (0.99 - 0.00)^2 + \\ (0.63 - 0.00)^2 \end{array} \right.$$



Backpropagation

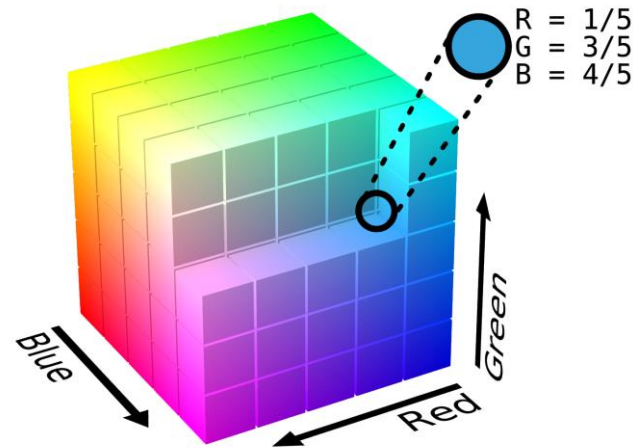




Word embeddings

Le parole o i caratteri non hanno la proprietà di essere valori continui come ad esempio i colori. Bisogna quindi trattarli come classi distinte, tipicamente per le NN si usa la codifica one-hot.

Classe	One-hot
rosso	00000001
verde	00000010
blu	00000100
giallo	00001000
viola	00010000
azzurro	00100000
nero	01000000
bianco	10000000



È possibile ottenere una rappresentazione continua delle parole?

$$W : \text{words} \rightarrow \mathbb{R}^n$$

$$W(\text{"cat"}) = (0.2, -0.4, 0.7, \dots)$$

$$W(\text{"mat"}) = (0.0, 0.6, -0.1, \dots)$$

Word embeddings

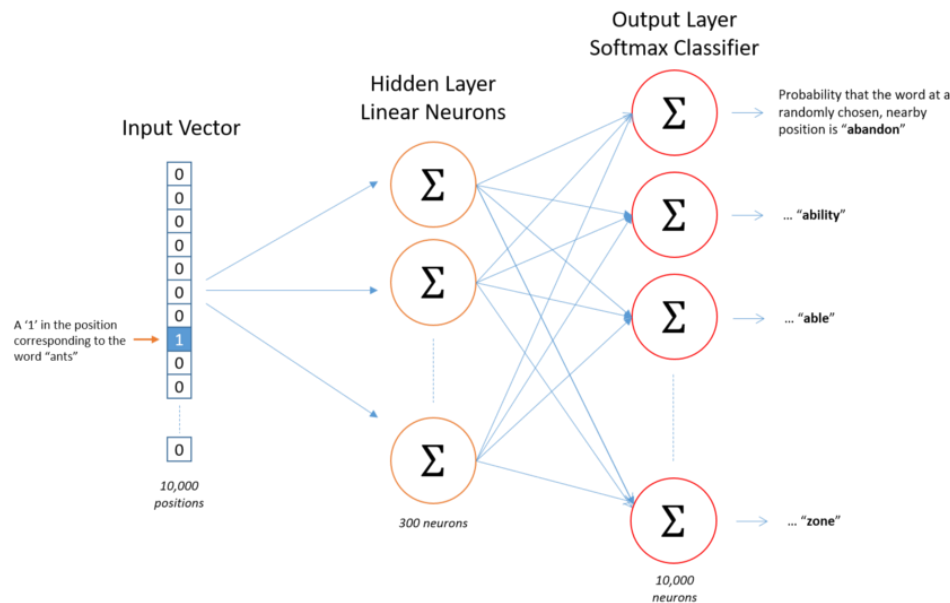
Il word embedding è una funzione parametrizzata che mappa le parole del linguaggio naturale in vettori di grandi dimensioni (100-800 valori). Alcuni esempi: [Word2vec](#), [GloVe](#), [FastText](#).

$$W : \text{words} \rightarrow \mathbb{R}^n$$

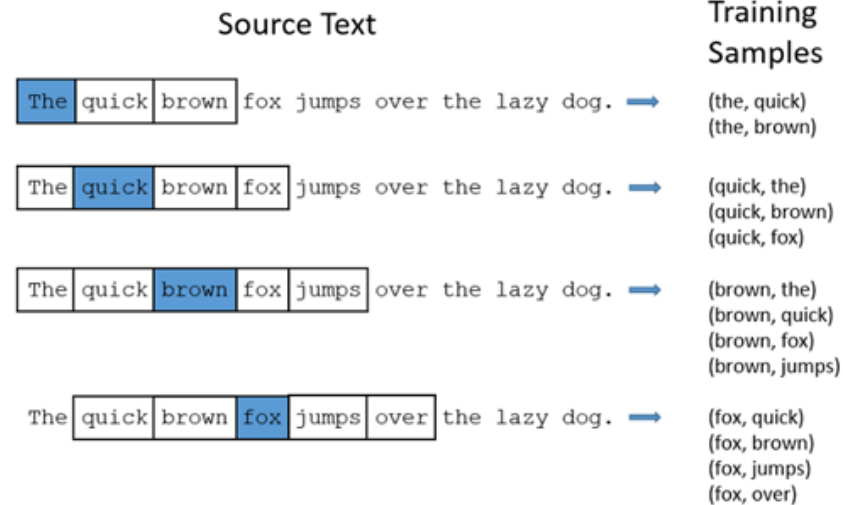
$$W(\text{"cat"}) = (0.2, -0.4, 0.7, \dots)$$

$$W(\text{"mat"}) = (0.0, 0.6, -0.1, \dots)$$

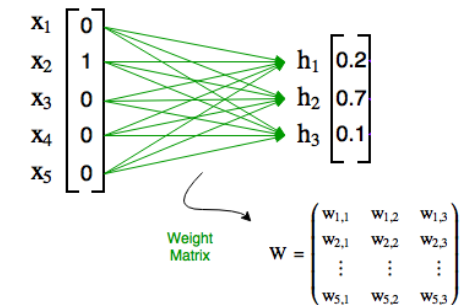
Model



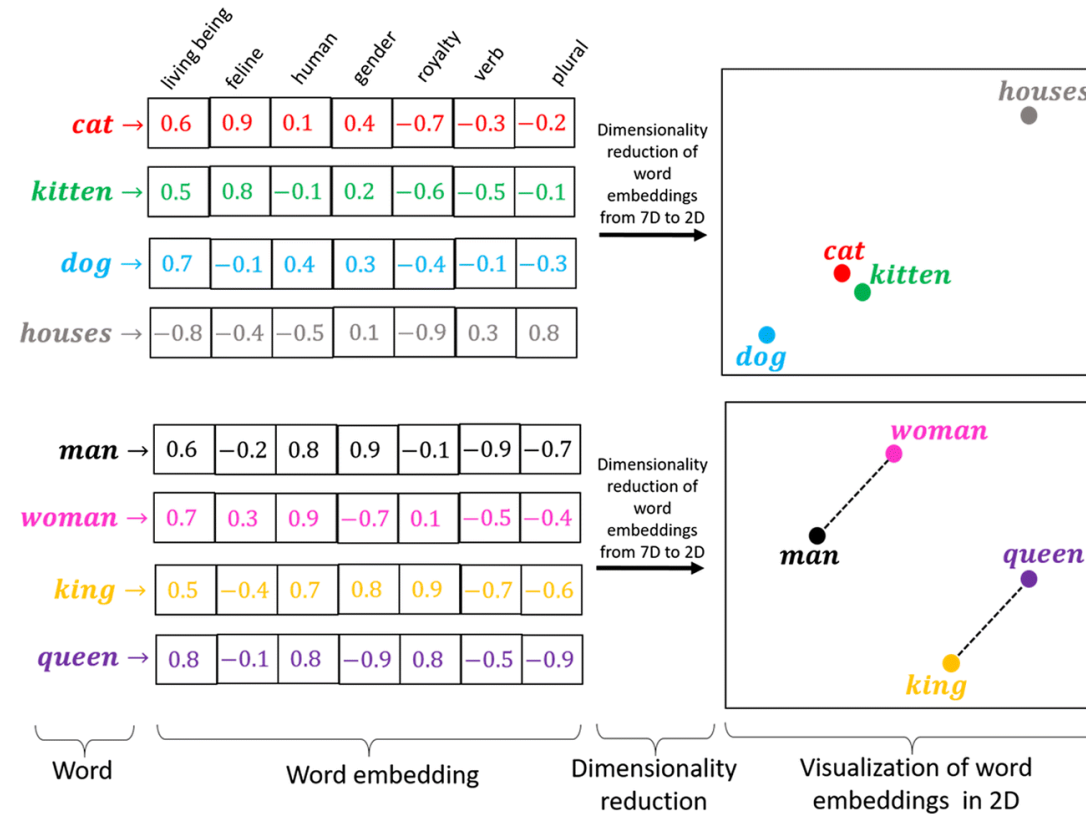
Training



Evaluation



Word embeddings



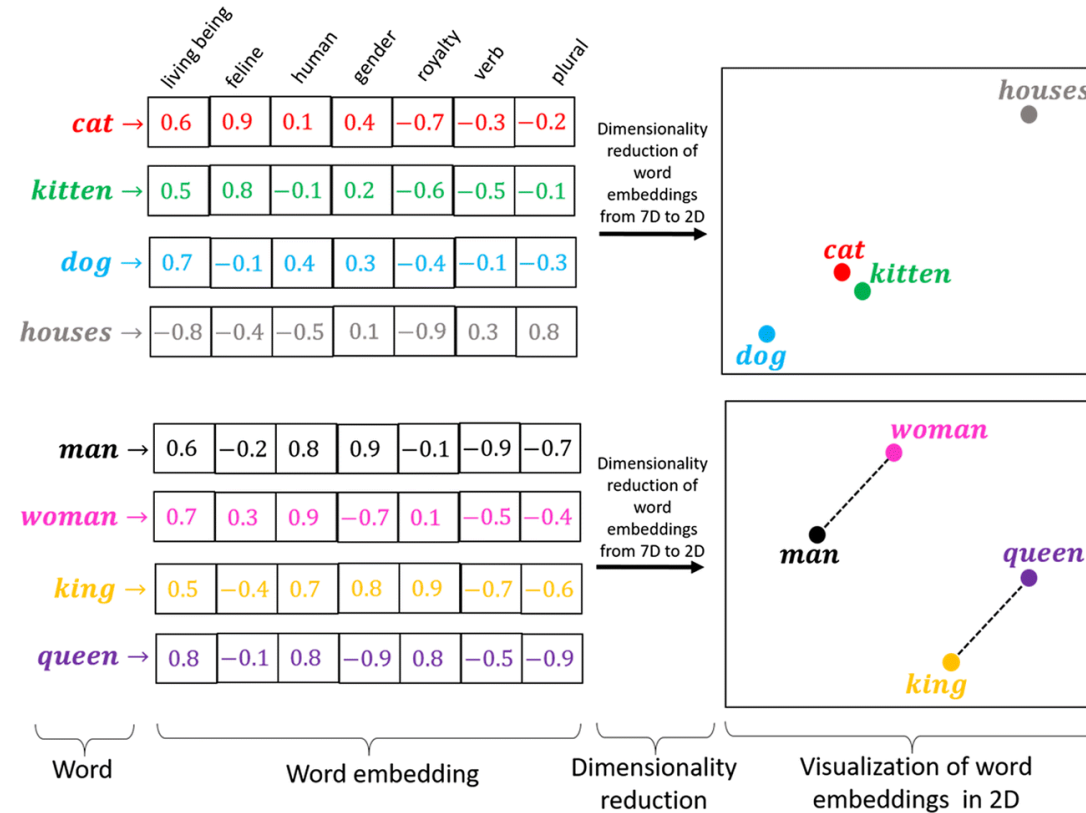
$$W : \text{words} \rightarrow \mathbb{R}^n$$

$$W(\text{"cat"}) = (0.2, -0.4, 0.7, \dots)$$

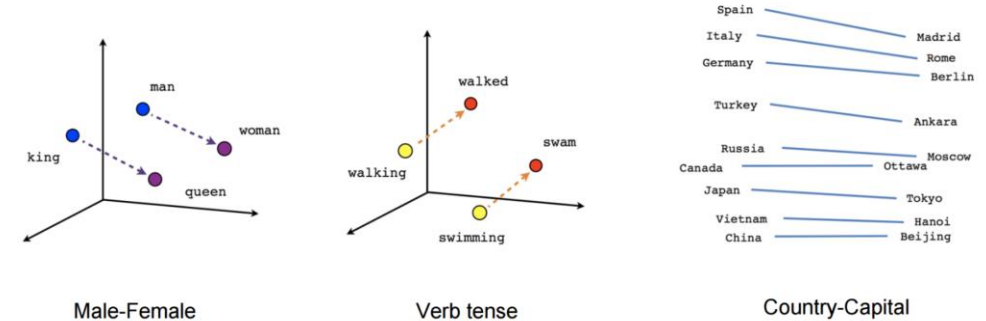
$$W(\text{"mat"}) = (0.0, 0.6, -0.1, \dots)$$

$$W_{\text{king}} - W_{\text{man}} + W_{\text{woman}} \sim W_{\text{queen}}$$

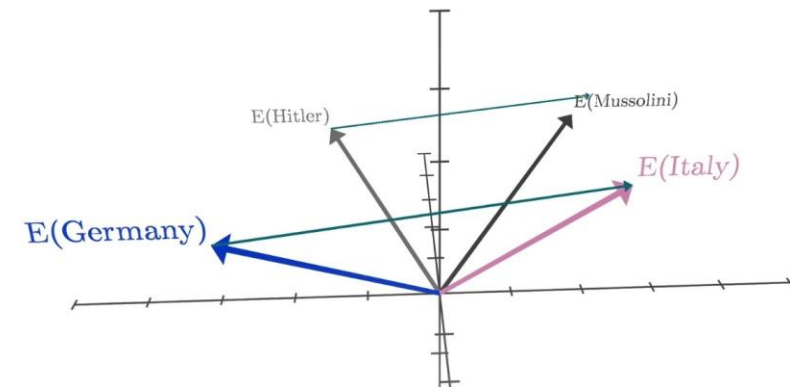
Word embeddings



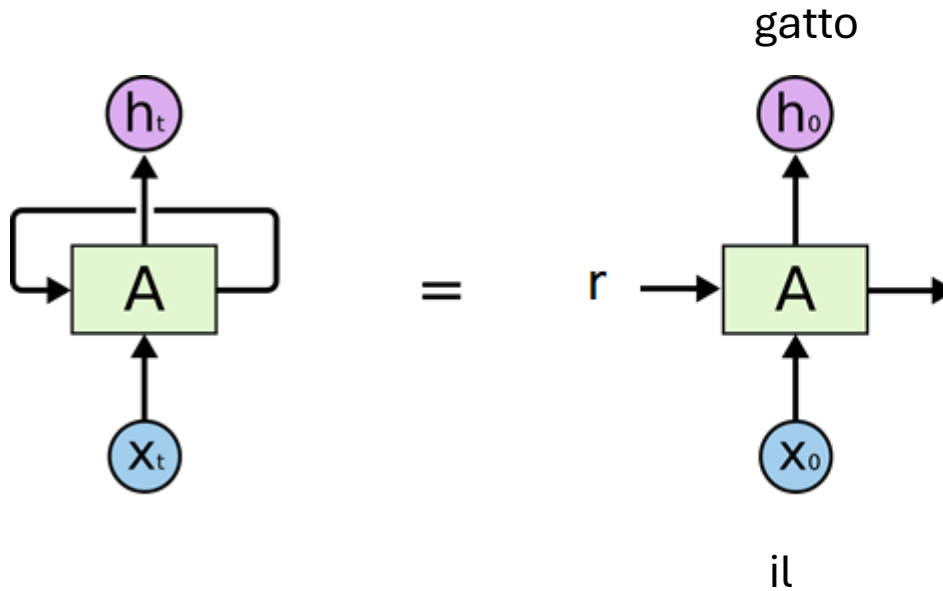
$$W_{\text{king}} - W_{\text{man}} + W_{\text{woman}} \sim W_{\text{queen}}$$



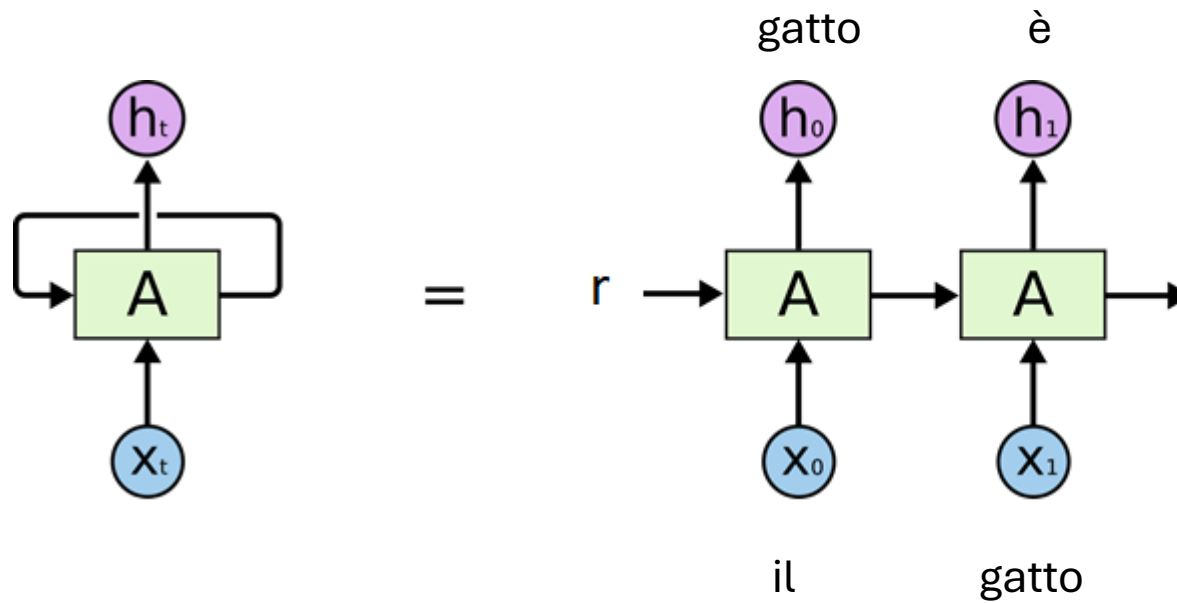
$$E(\text{Hitler}) + E(\text{Italy}) - E(\text{Germany}) \approx E(\text{Mussolini})$$



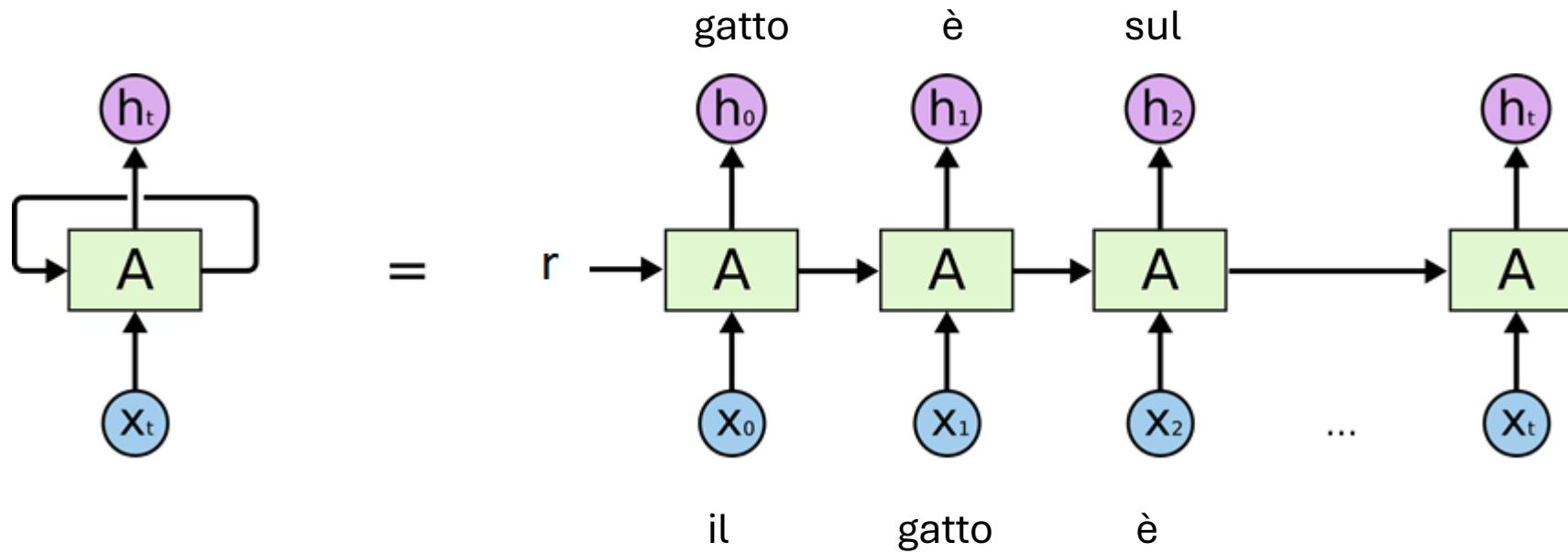
RNN: Recurrent Neural Network



RNN: Recurrent Neural Network



RNN: Recurrent Neural Network



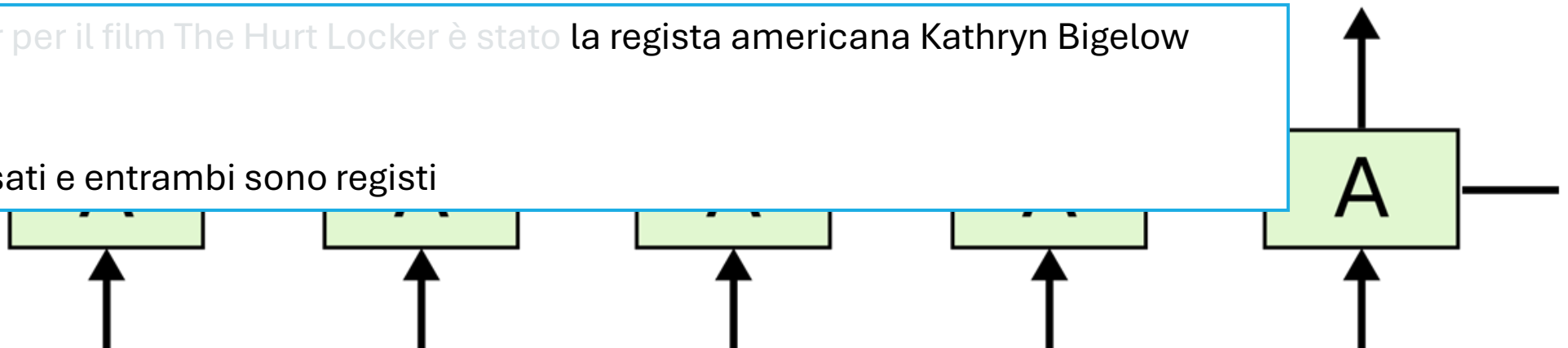
Autoregressione e allucinazioni

A vincere l'Oscar per il film The Hurt Locker è stato il regista americano James

Posizione / Scelta	Parola 1	Parola 2	Parola 3	Parola 4	Parola 5
1° Candidato	il (75%)	regista (90%)	americano (65%)	James (45%)	Cameron (95%)
2° Candidato	un (15%)	cineasta (5%)	statunitense (25%)	Martin (20%)	Mangold (3%)
3° Candidato	nominato (5%)	talento (3%)	di (5%)	Steven (15%)	Gunn (1%)
4° Candidato	la (3%)	brillante (1.5%)	visionario (4%)	Kathryn (10%)	Ivory (0.5%)
5° Candidato	proprio (2%)	produttore (0.5%)	indipendente (1%)	Quentin (10%)	Franco (0.5%)

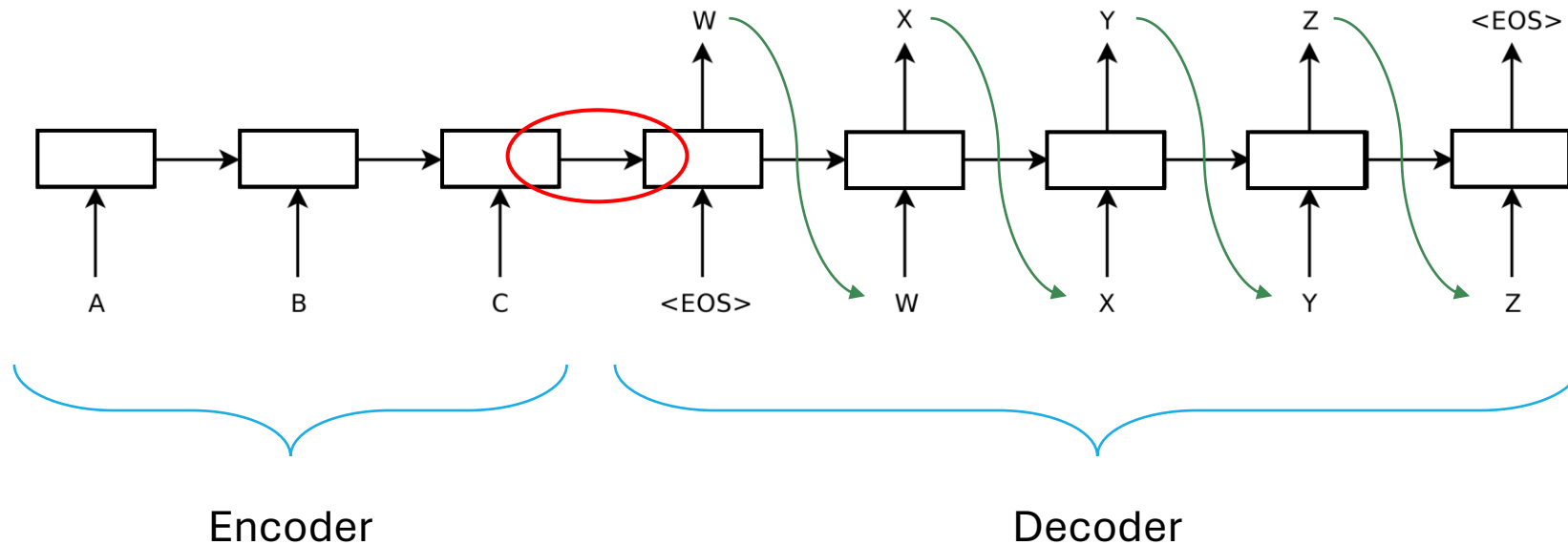
A vincere l'Oscar per il film The Hurt Locker è stato la regista americana Kathryn Bigelow

* Sono stati sposati e entrambi sono registi



Seq2seq

- Difficile da allenare
 - Difficile da parallelizzare
 - Gradiente molto lungo (oltre 100 layer)
- Limitate capacità di capire il contesto passato.

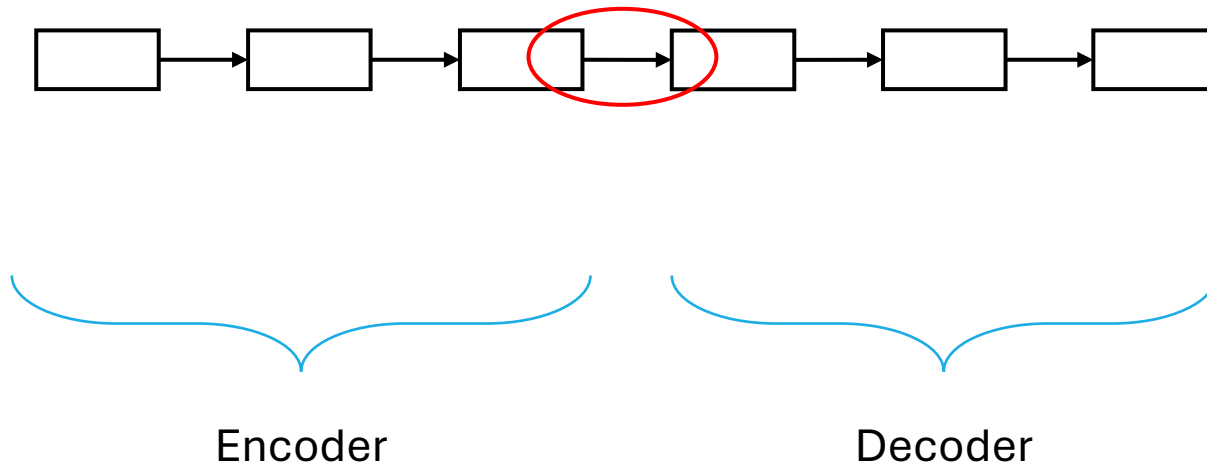


<https://arxiv.org/abs/1409.3215>

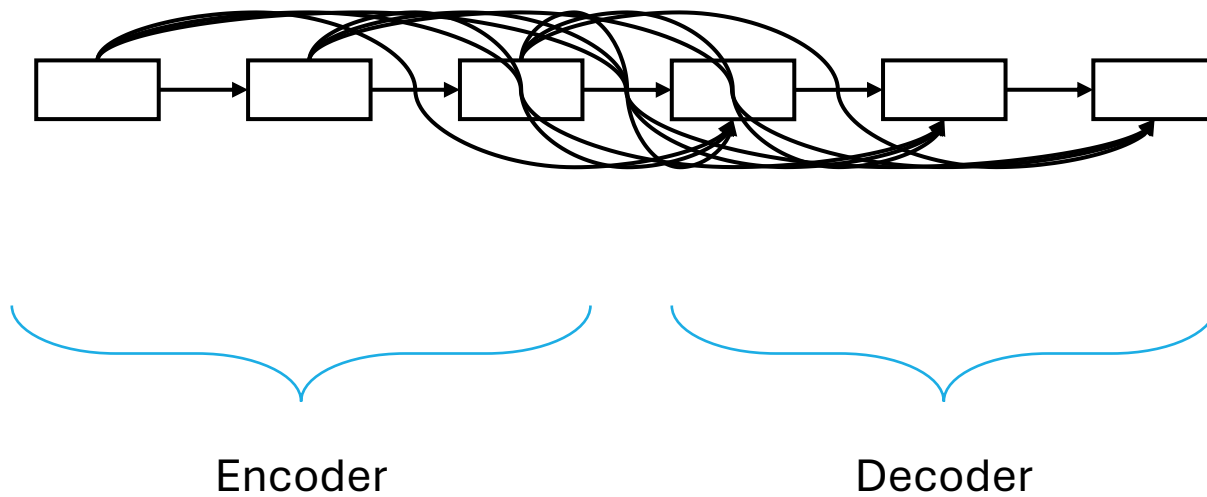
<https://www.youtube.com/watch?v=quoGRI-1l0A> C5W3L08 Attention Model

<https://lilianweng.github.io/lil-log/2018/06/24/attention-attention.html>

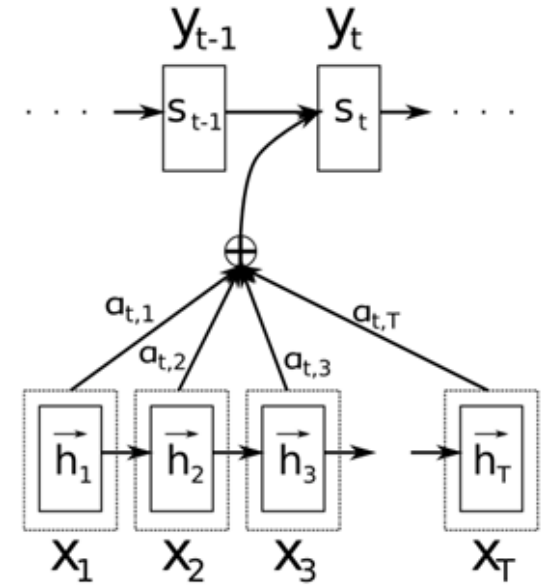
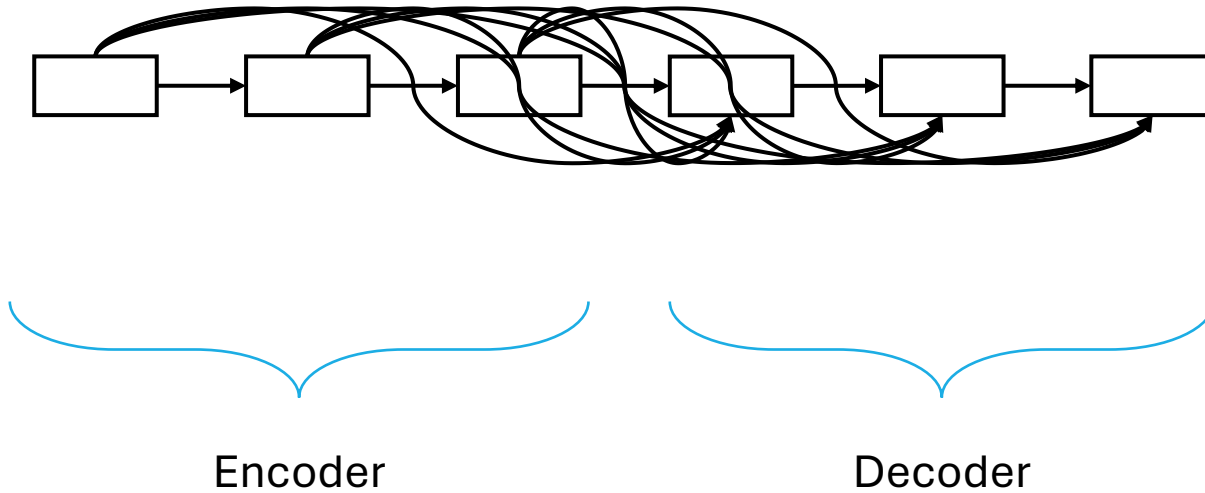
Seq2seq



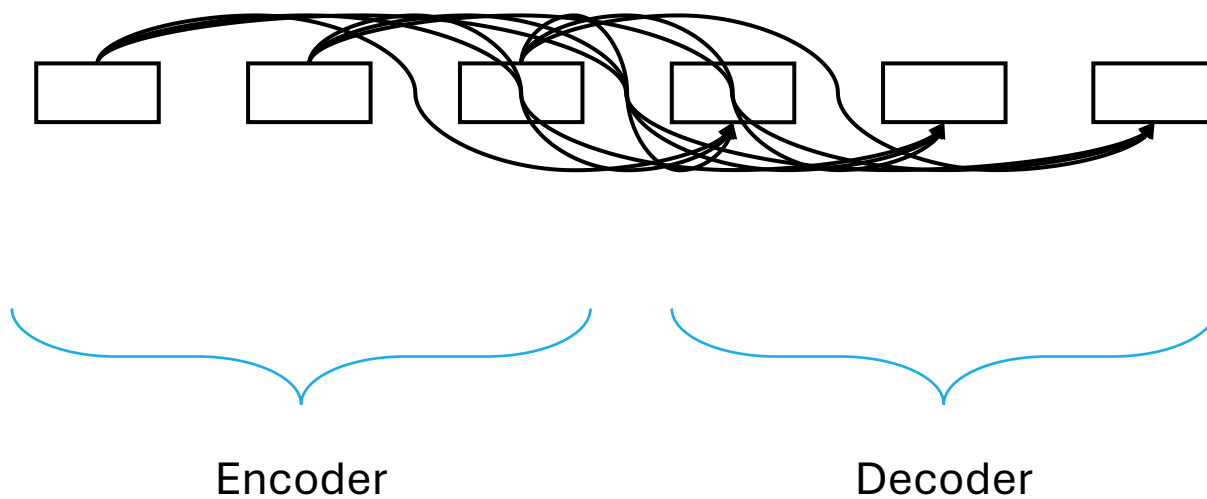
Seq2seq



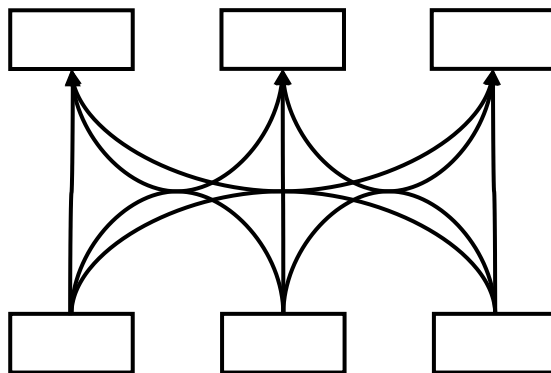
Attention in RNN



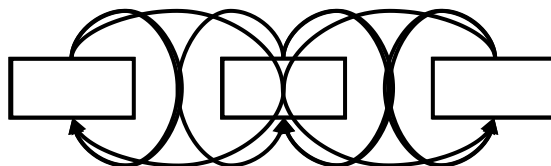
Attention

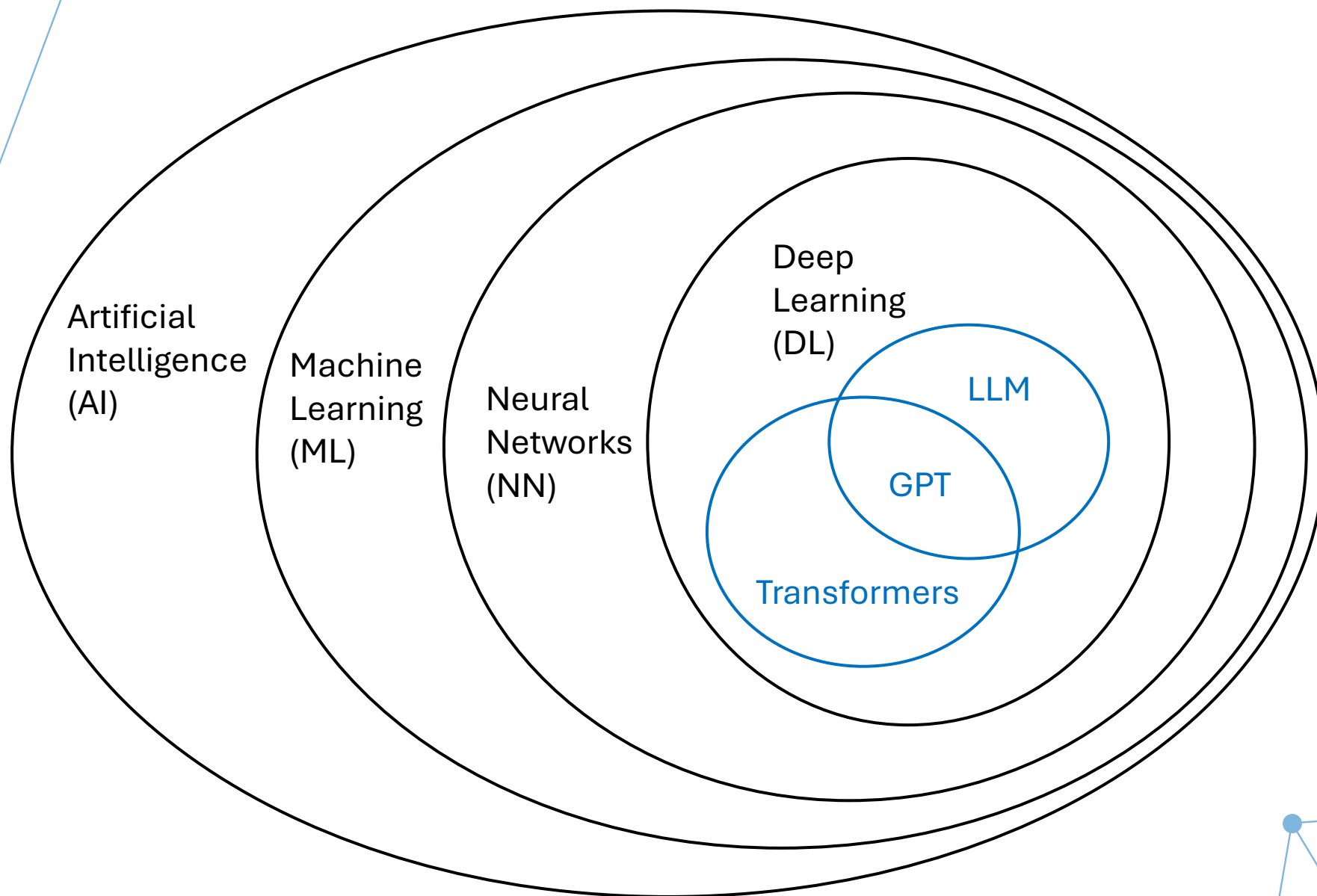


Attention



Self Attention





Attention Is All You Need

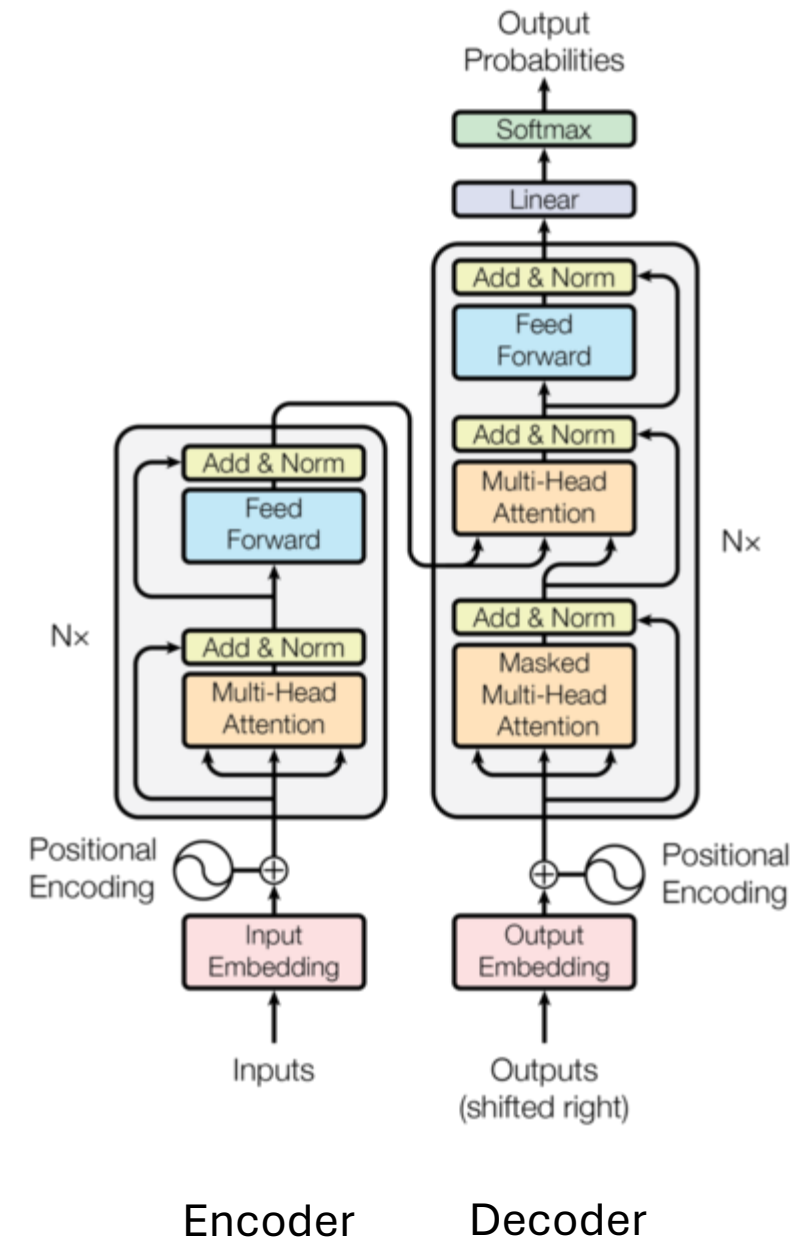
<https://arxiv.org/abs/1706.03762> (2017)

Inizialmente pensato per traduzioni, seq2seq.

Rimane solo l'attenzione e rimuove la parte RNN o CNN.
Collegamenti tutti verso tutti (self attention).

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	



Attention Is All You Need

Vantaggi:

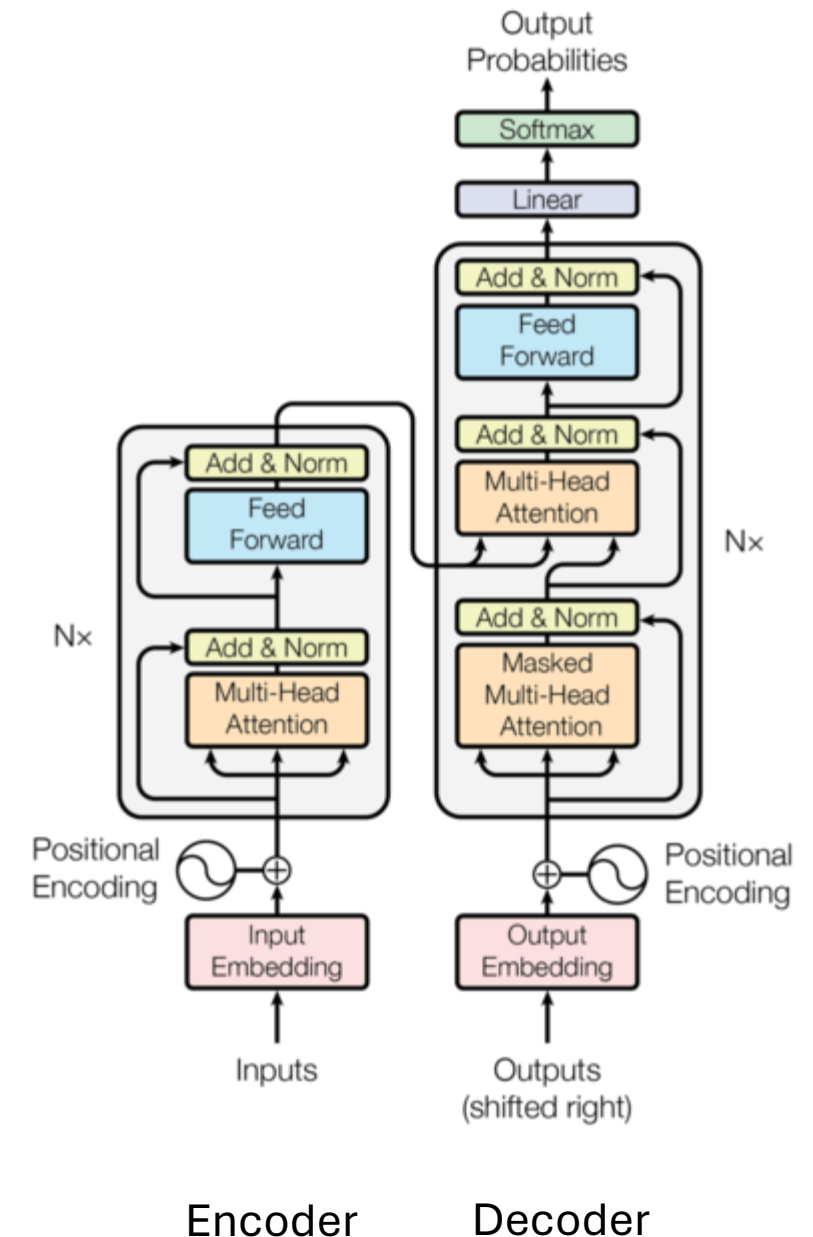
- Più parallelizzabile (-> più veloce -> più dati in allenamento)
- Risultati migliori (impara dipendenze lontane)
- Computazione lineare nella profondità della rete

Svantaggi:

- Dimensione fissa dell'input
- Computazione esponenziale nella lunghezza dell'input

Table 1: Maximum path lengths, per-layer complexity and minimum number of sequential operations for different layer types. n is the sequence length, d is the representation dimension, k is the kernel size of convolutions and r the size of the neighborhood in restricted self-attention.

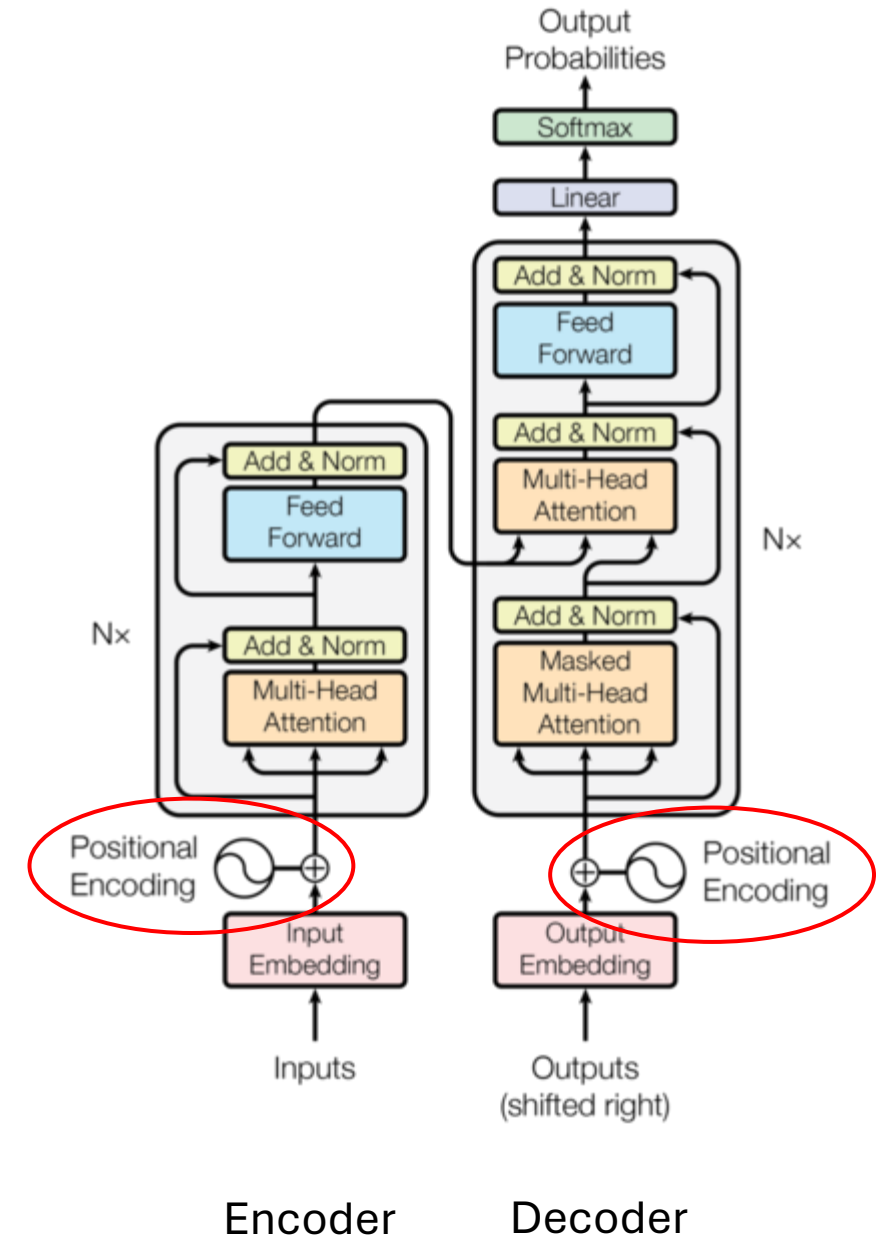
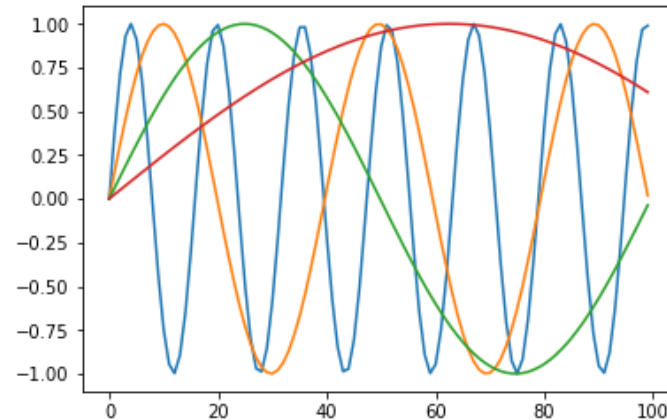
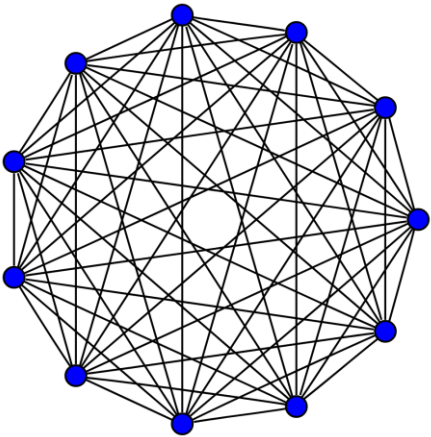
Layer Type	Complexity per Layer	Sequential Operations	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$



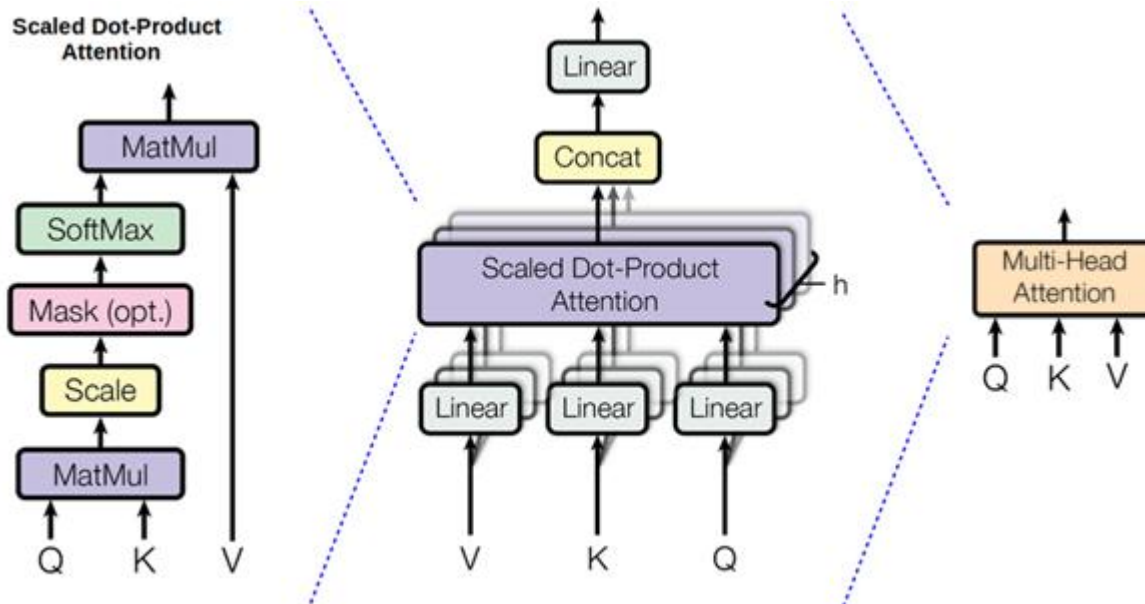
Attention Is All You Need

Abbiamo perso la causalità rimuovendo i collegamenti RNN, come si può informare il modello sulla posizione dei token?

Positional encoding

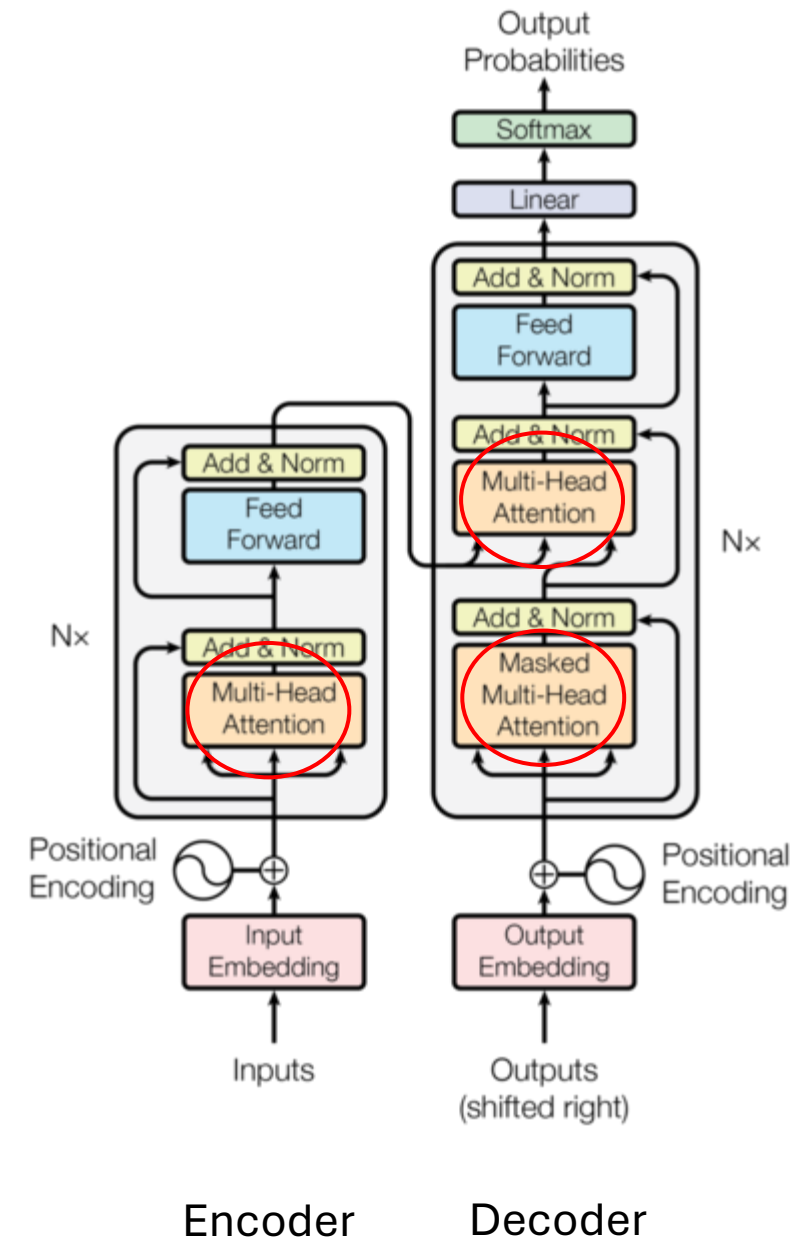


Attention Is All You Need



Q, K, V sono ottenuti dagli embedding con i nuovi pesi W_q , W_k , W_v .

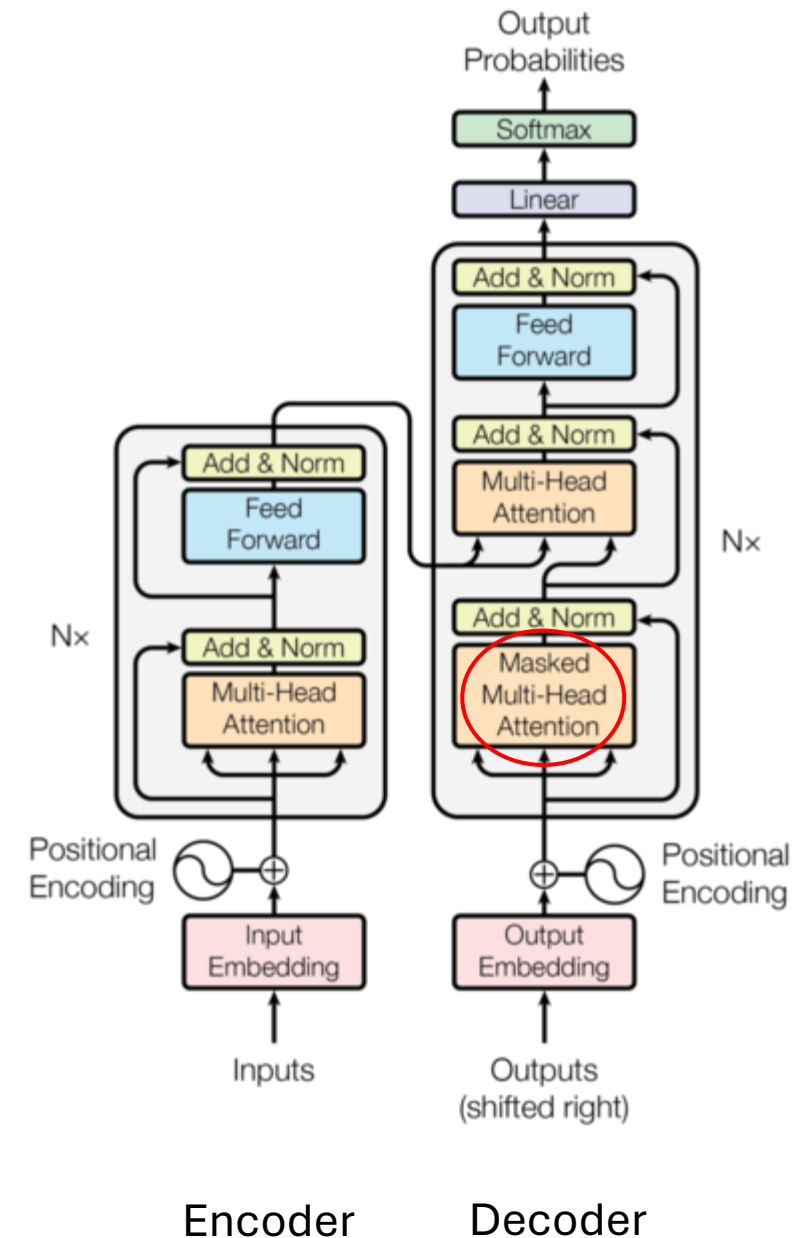
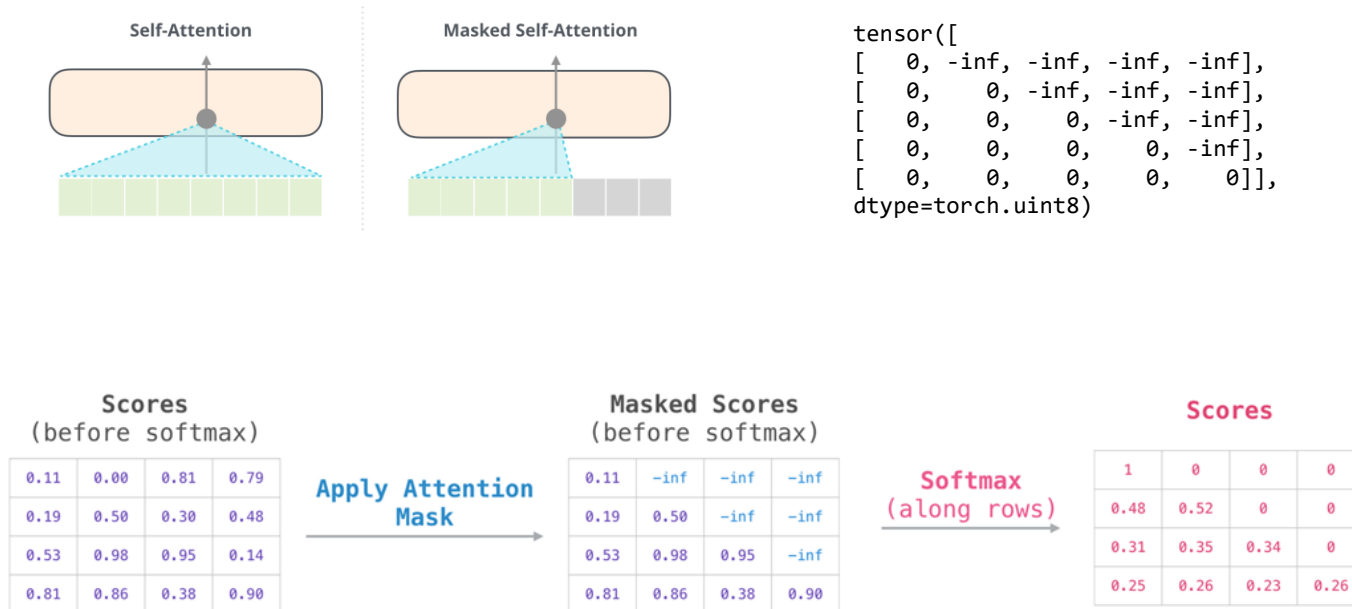
$$\text{Attention}(Q, K, V) = \text{softmax}_k\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Attention Is All You Need

Masked multi-head attention:

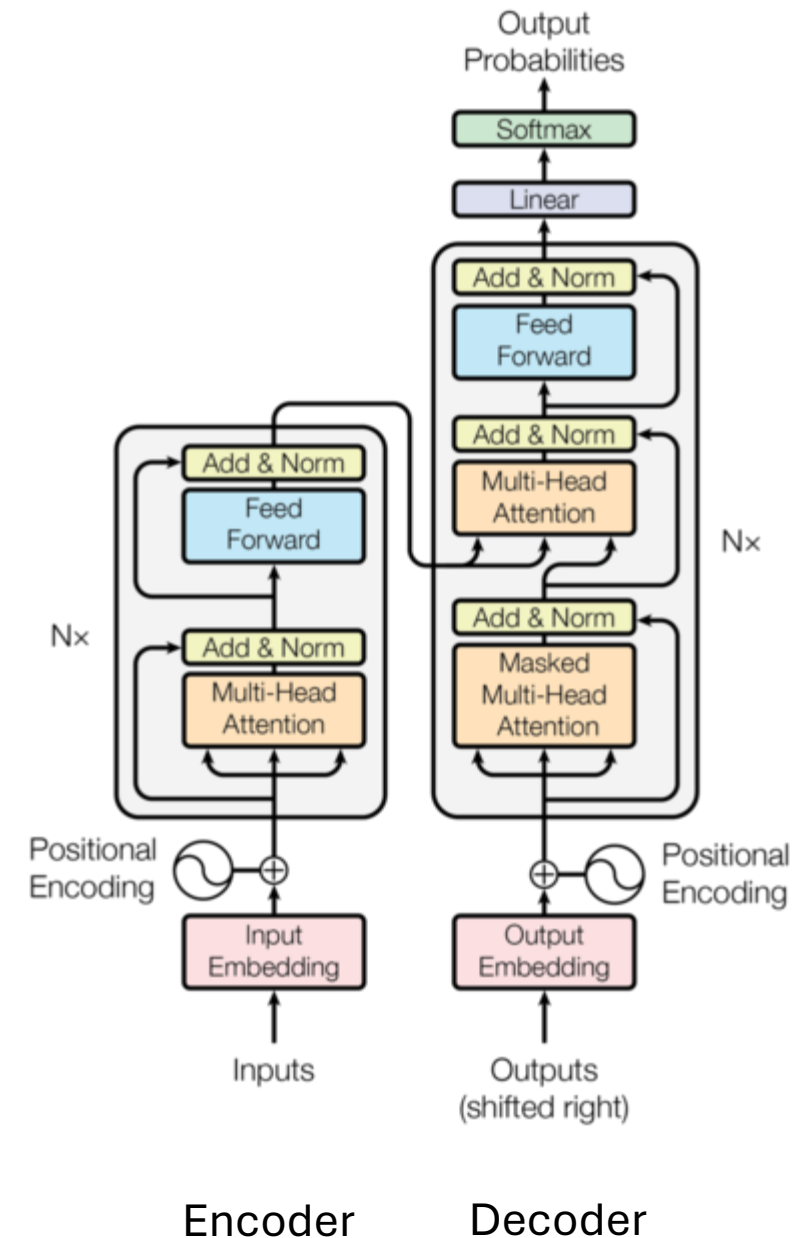
ciascun token non vede i suoi token successivi.

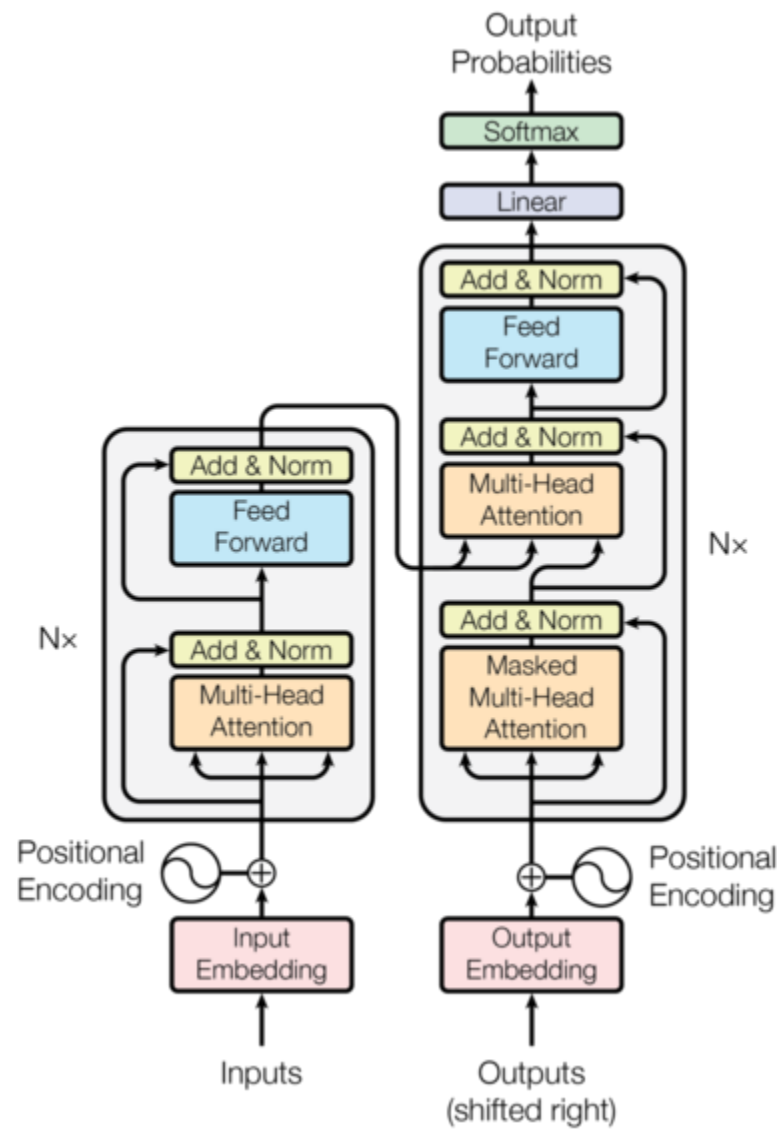


Attention Is All You Need

Maggiori dettagli e spiegazione grafica:

<https://jalammar.github.io/illustrated-transformer/>





BERT & GPT 2018



BERT



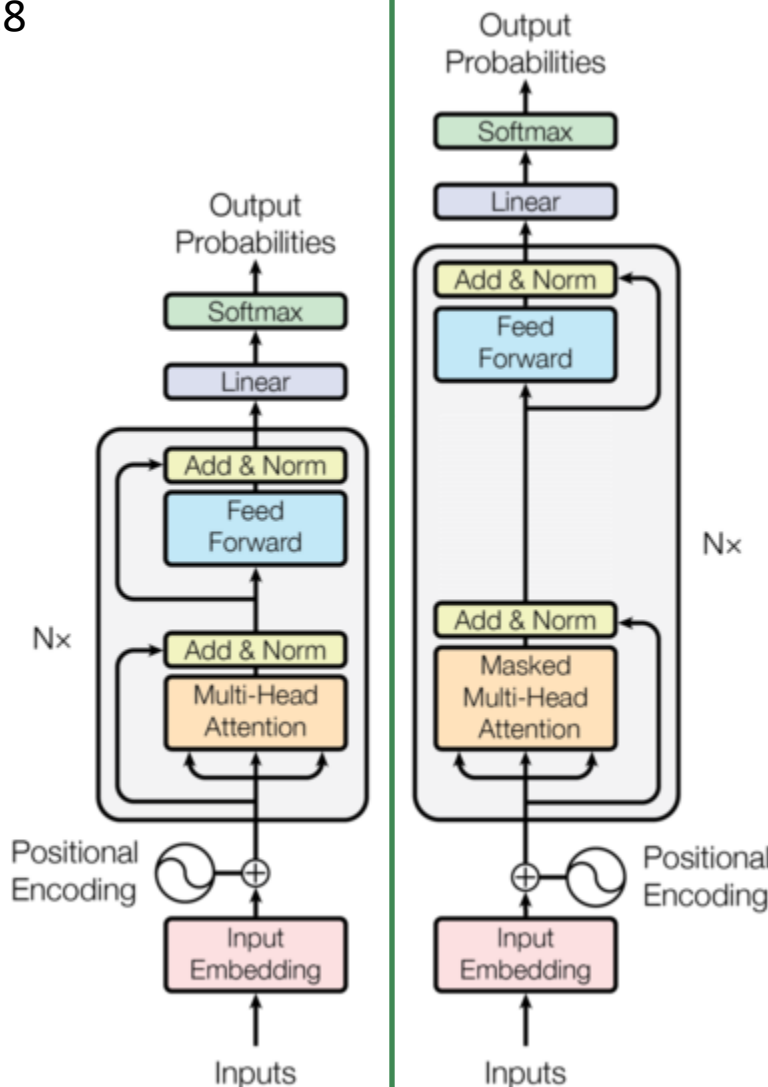
Parte encoder di Transformer

Scopo:

Applicare il transfer learning al modello Transformer.

SOTA:

GLUE, SQuAD v1.1, SQuAD v2.0



GPT

OpenAI

Parte decoder di Transformer

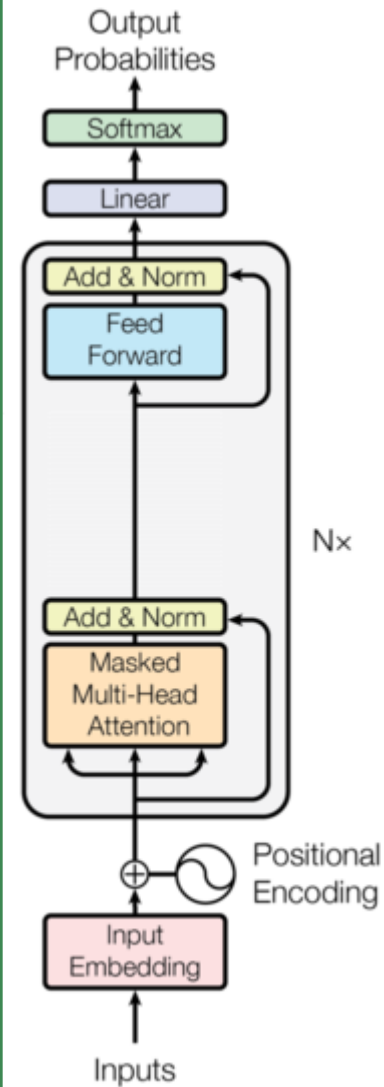
Scopo:

Ottenere il miglior modello linguistico causale.

SOTA:

zero-shot tasks

GPT



OpenAI

Parte decoder di Transformer

Scopo:
Ottenere il miglior modello
linguistico causale.

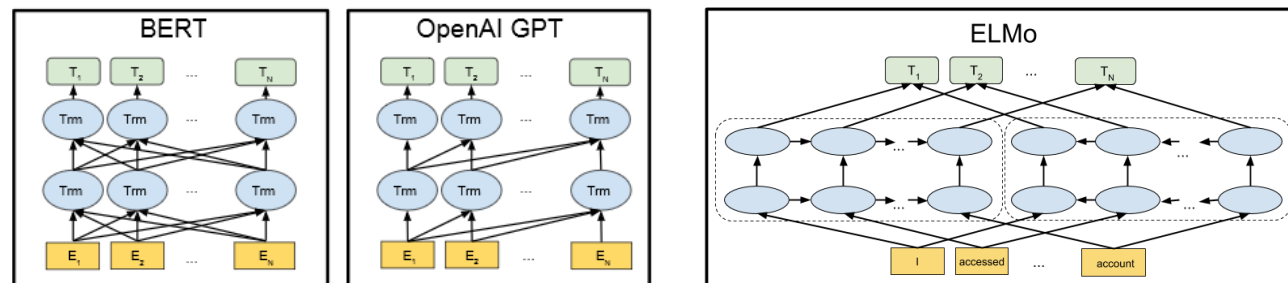
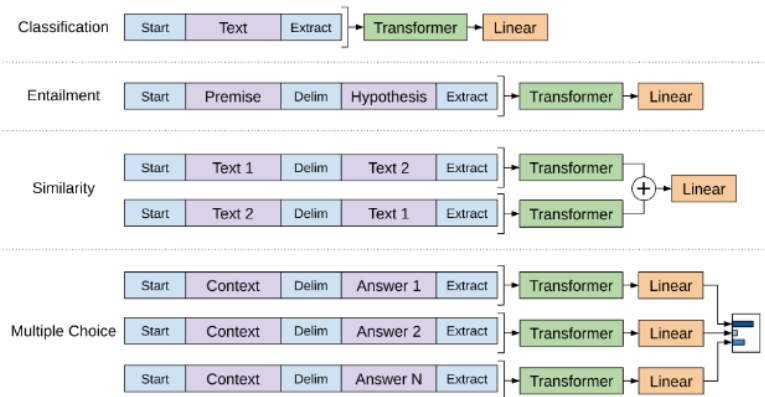
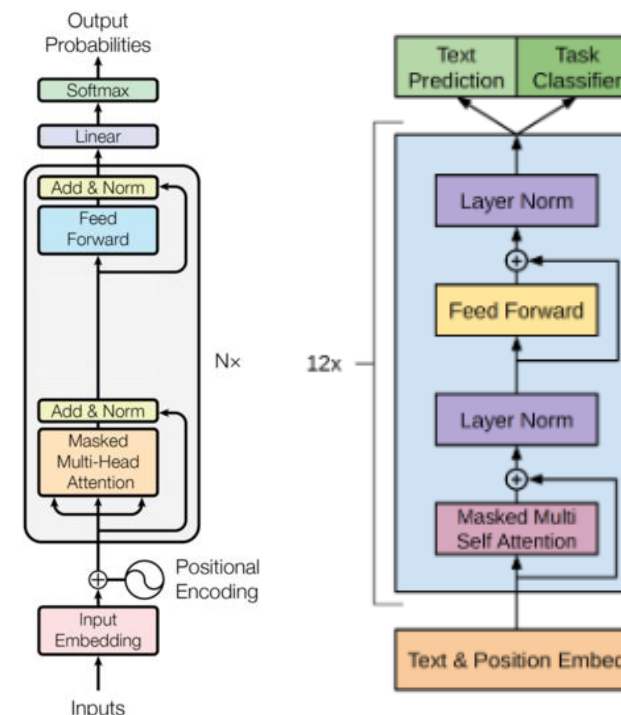
SOTA:
zero-shot tasks

GPT

Generative Pre-trained Transformer

https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf

In sostanza la parte decoder di “Attention is all you need”, dedicata a creare un modello linguistico.



GPT

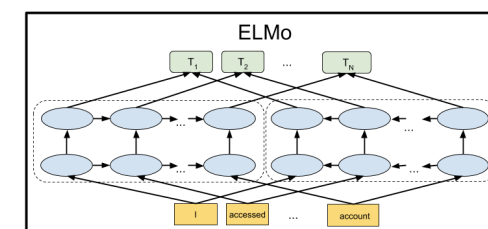
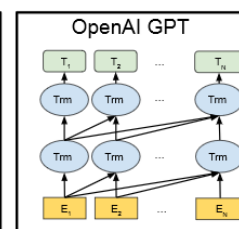
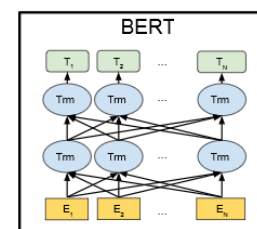
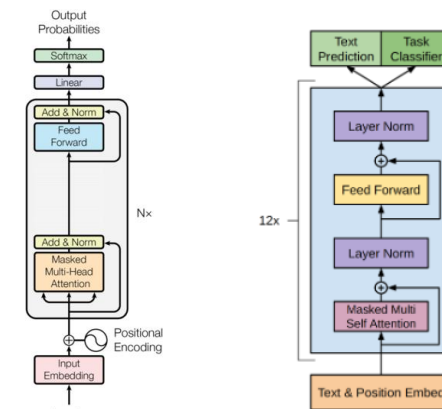
Giugno 2018

Corpus: ~5 miliardi di token

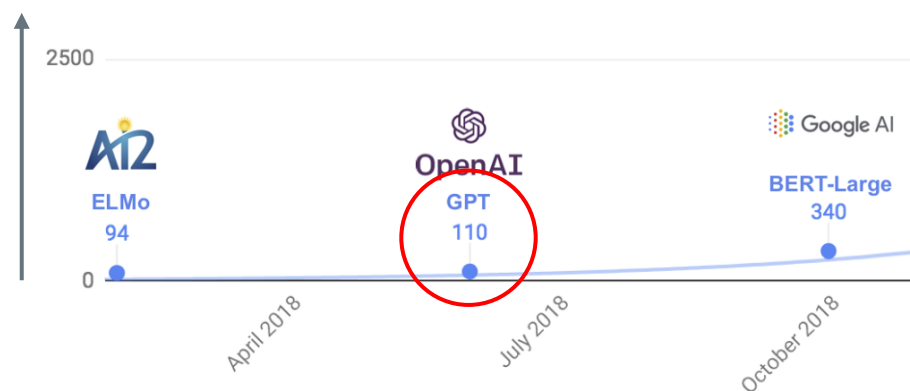
Modello: 117 milioni di parametri

Illustrated gpt2, like illustrated transformer:

<https://jalammar.github.io/illustrated-gpt2/>



Milioni di parametri



GPT2

Febbraio 2019

<https://openai.com/blog/better-language-models/>

Quanto un modello transformer può imparare?

Modello: 1.5 miliardi di parametri

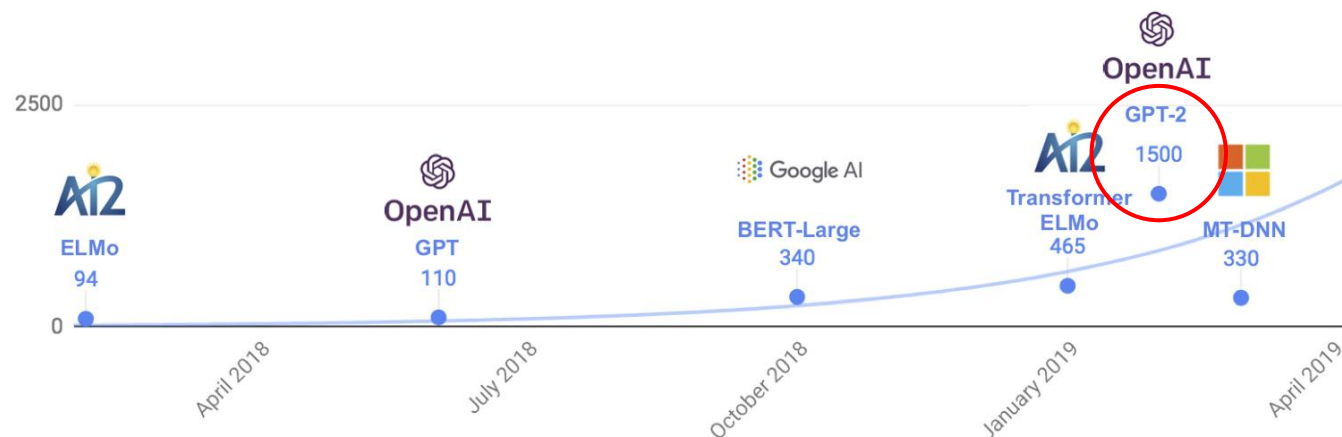
Corpus: 10 miliardi di token (40GB)

Language Models are Unsupervised Multitask Learners

Alec Radford^{*1} Jeffrey Wu^{*1} Rewon Child¹ David Luan¹ Dario Amodei^{**1} Ilya Sutskever^{**1}

3.6. Summarization

We test GPT-2's ability to perform summarization on the CNN and Daily Mail dataset (Nallapati et al., 2016). To induce summarization behavior we add the text TL;DR: after the article and generate 100 tokens with Top- k random sampling (Fan et al., 2018) with $k = 2$ which reduces repetition and encourages more abstractive summaries than greedy de-



GPT2

Febbraio 2019

<https://openai.com/blog/better-language-models/>

Quanto un modello transformer può imparare?

Modello: 1.5 miliardi di parametri

Corpus: 10 miliardi di token (40GB)



The Guardian

News Opinion Sport Culture Lifestyle

World UK Climate crisis Ukraine Environment Science Global development Football Tech Business

AI (artificial intelligence)

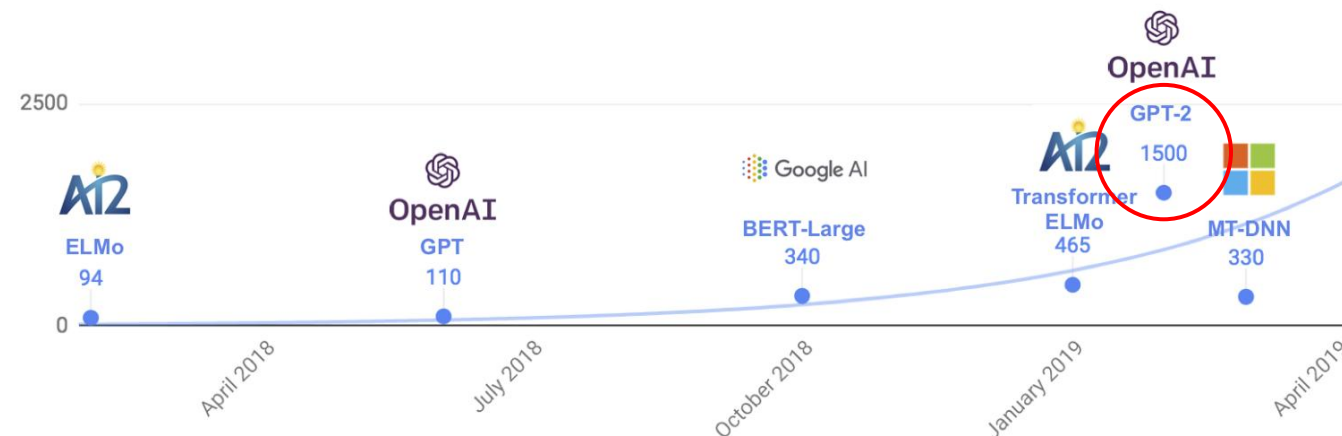
This article is more than 7 years old

New AI fake text generator may be too dangerous to release, say creators

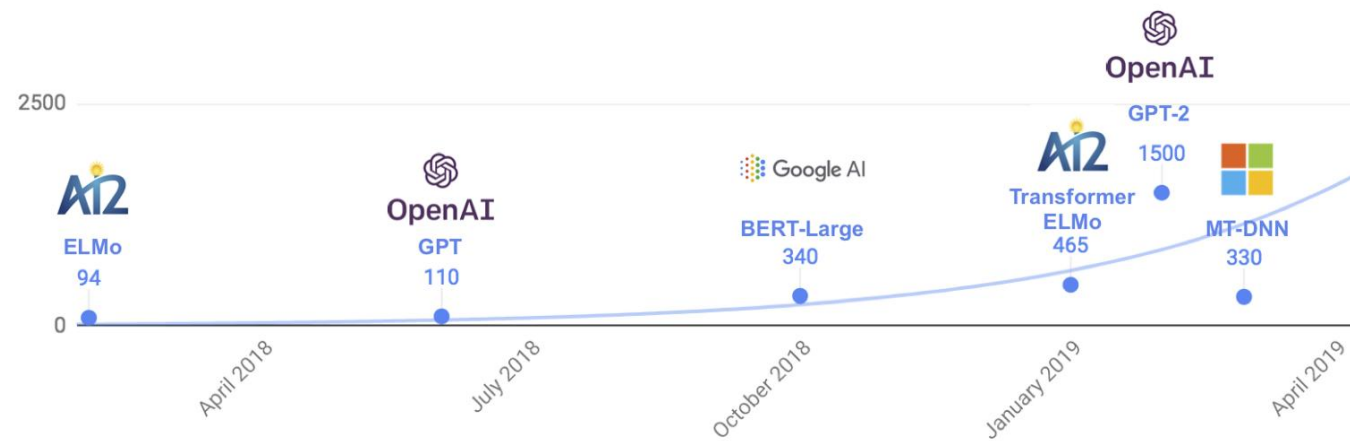
The Elon Musk-backed nonprofit company OpenAI declines to release research publicly for fear of misuse

Quick Guide
Here's what happened when we asked OpenAI to write about itself

We gave OpenAI's AI the first two paragraphs of this story to see what it would write about itself. Here's the story it went with. To be clear: none of what follows is true.



GPT3?



GPT3?



GPT3

Maggio 2020

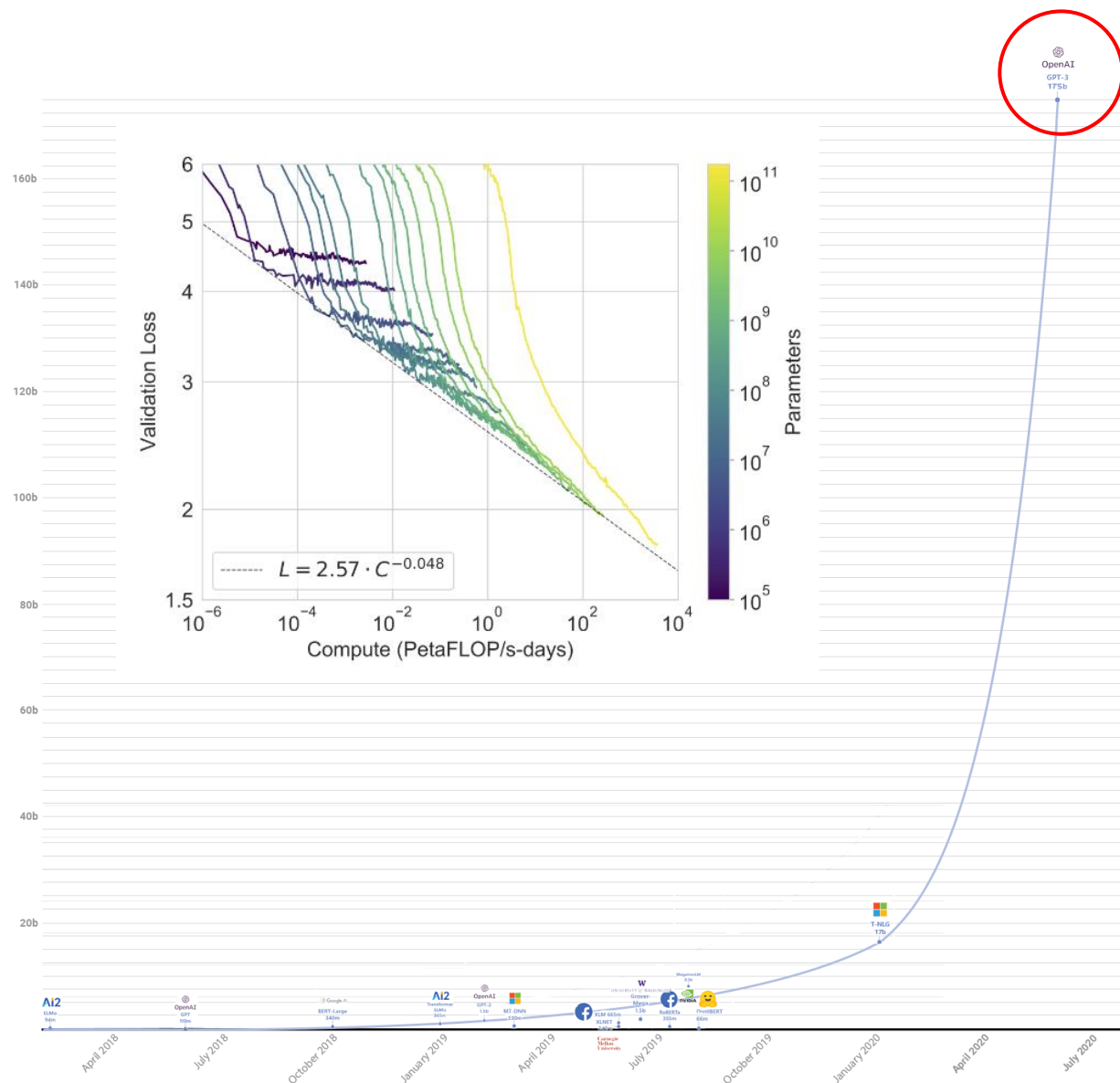
«Language Models are Few-Shot Learners»

<https://arxiv.org/abs/2005.14165>

Modello: 175 miliardi di parametri

Corpus: 499 miliardi di token (circa 45TB)

Costo: ~\$4,600,000 per riprodurre il modello di solo computazione (stima)



GPT3

«Language Models are Few-Shot Learners»

The three settings we explore for in-context learning

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

Few-shot

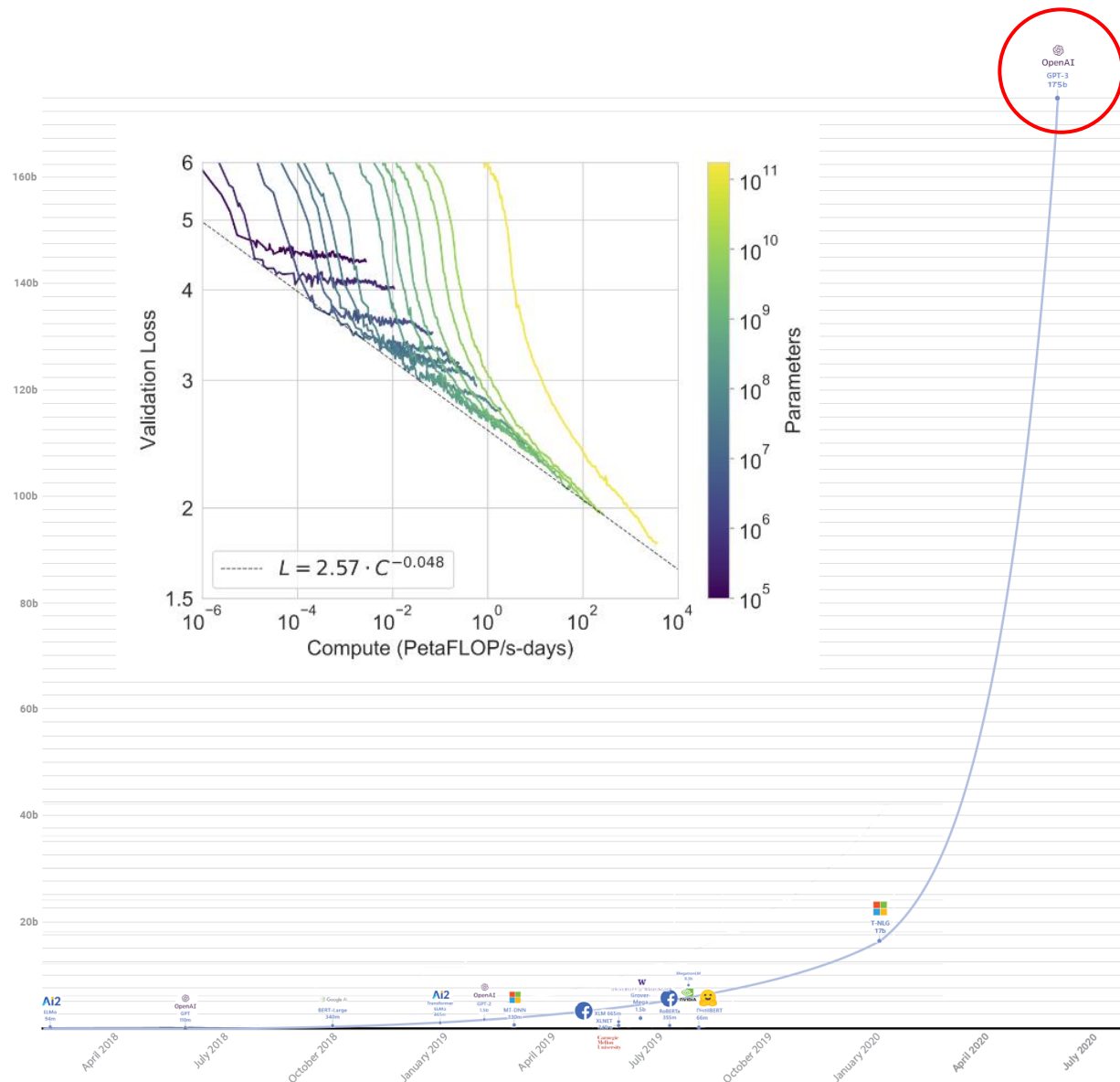
In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée
4 plush girafe => girafe peluche
5 cheese => ..... ← prompt
```

Traditional fine-tuning (not used for GPT-3)

Fine-tuning

The model is trained via repeated gradient updates using a large corpus of example tasks.



ChatGPT

Novembre 2022

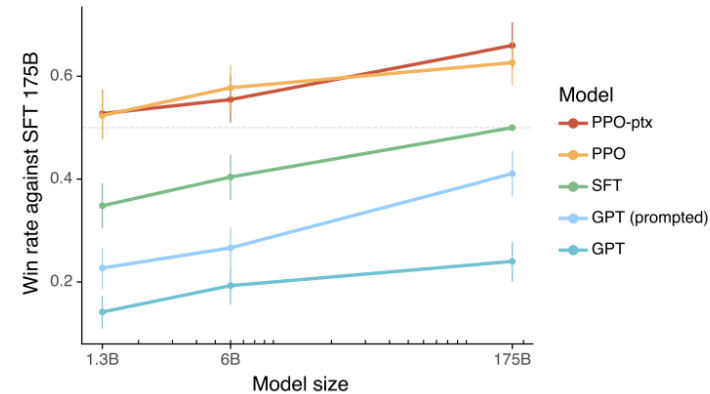
<https://arxiv.org/pdf/2203.02155>

Instruction tuning su GPT3 (InstructGPT)

Da completatore ad assistente

```
<|im_start|>user\nExplain the moon landing to a 6 year old in a few sentences.\n<|im_end|>\n<|im_start|>assistant\nPeople went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.
```

Training language models to follow instructions with human feedback



Prompt

Explain the moon landing to a 6 year old in a few sentences.

Completion

GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT

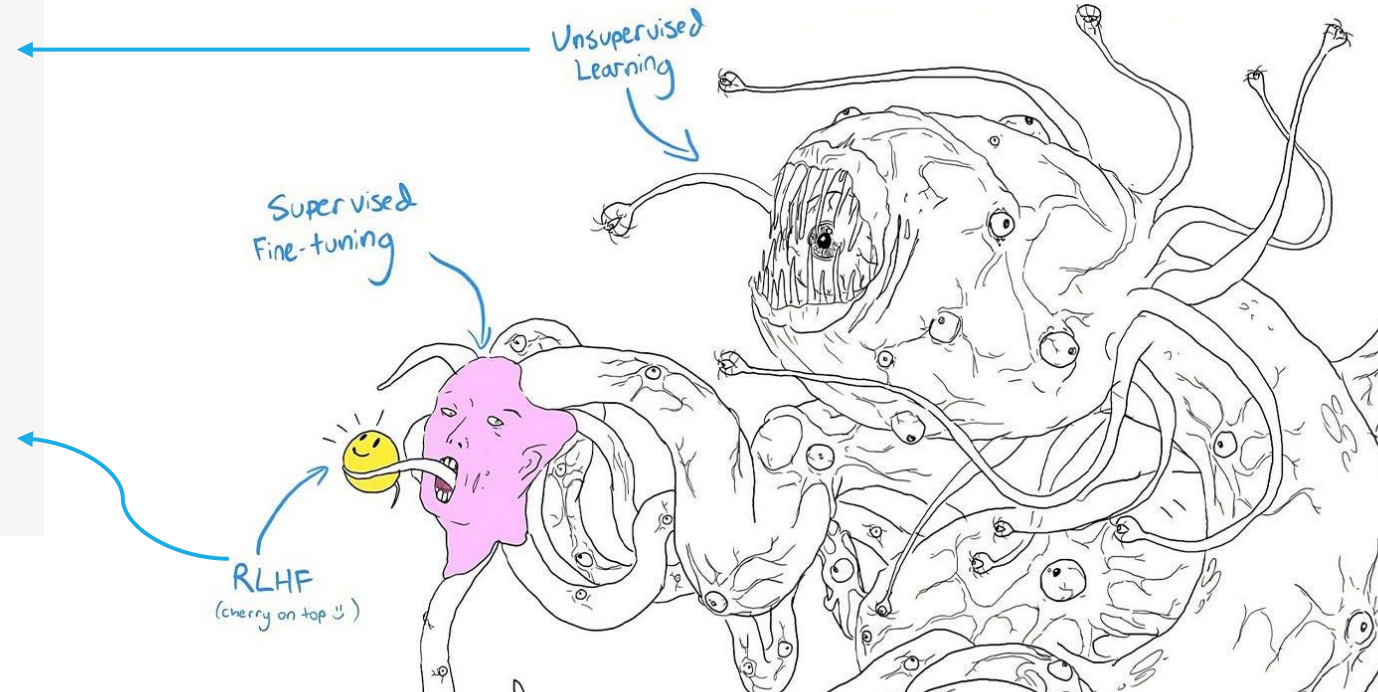
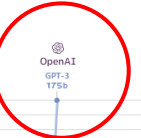
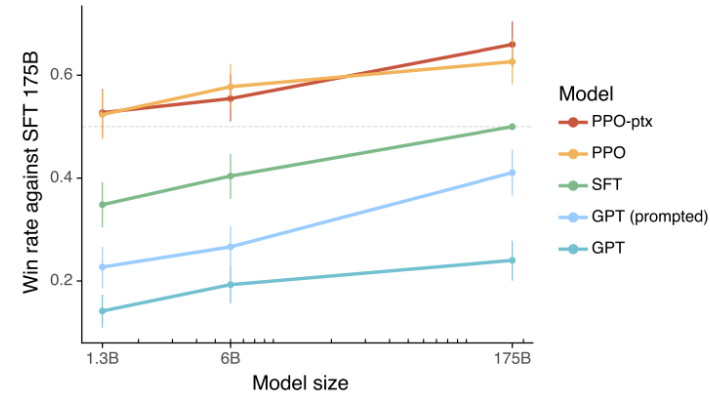
People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.



ChatGPT

Novembre 2022

Training language models to follow instructions with human feedback



Prompt

Explain the moon landing to a 6 year old in a few sentences.

Completion

GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

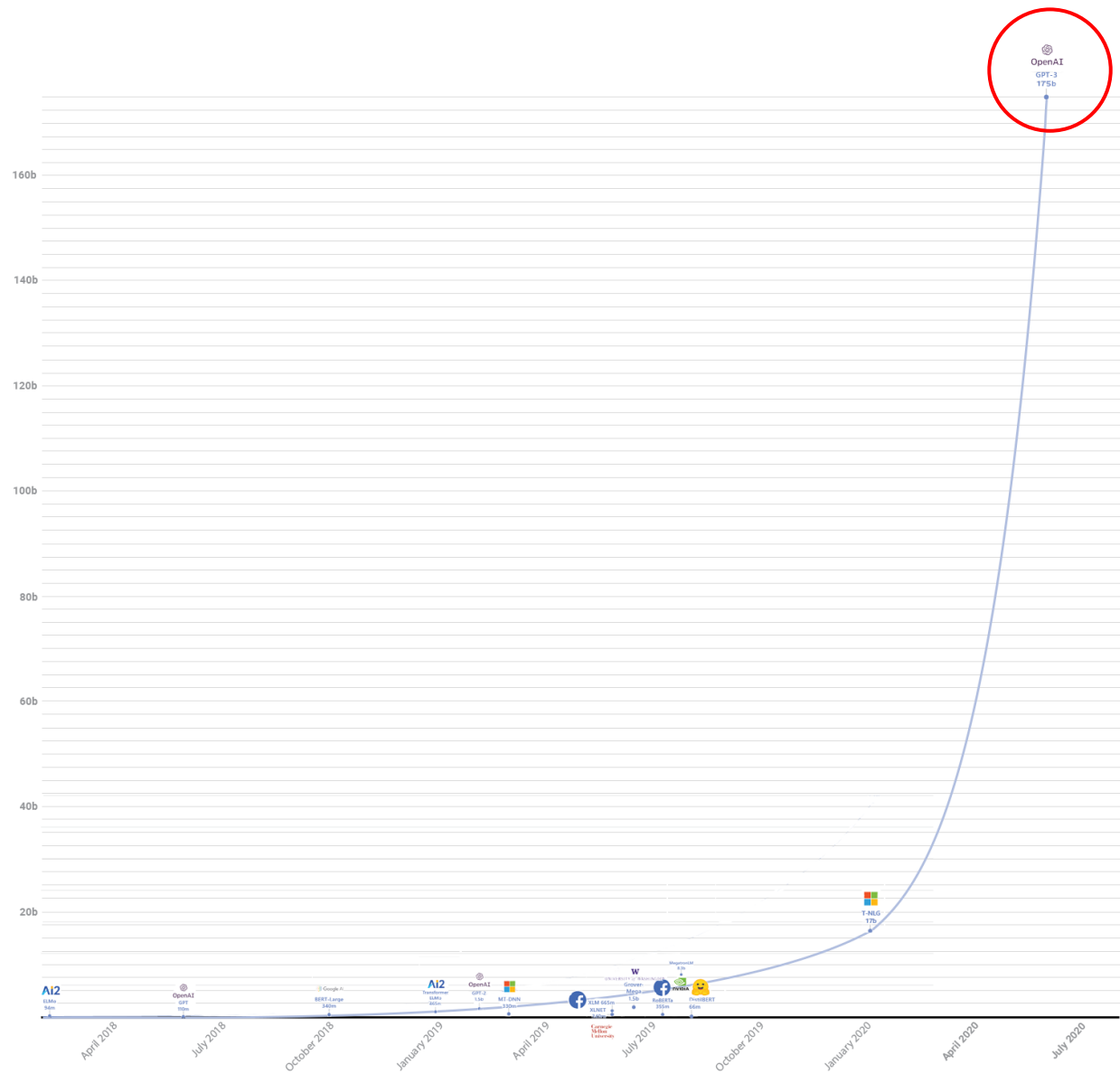
Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

GPT3



GPT3



GPT4

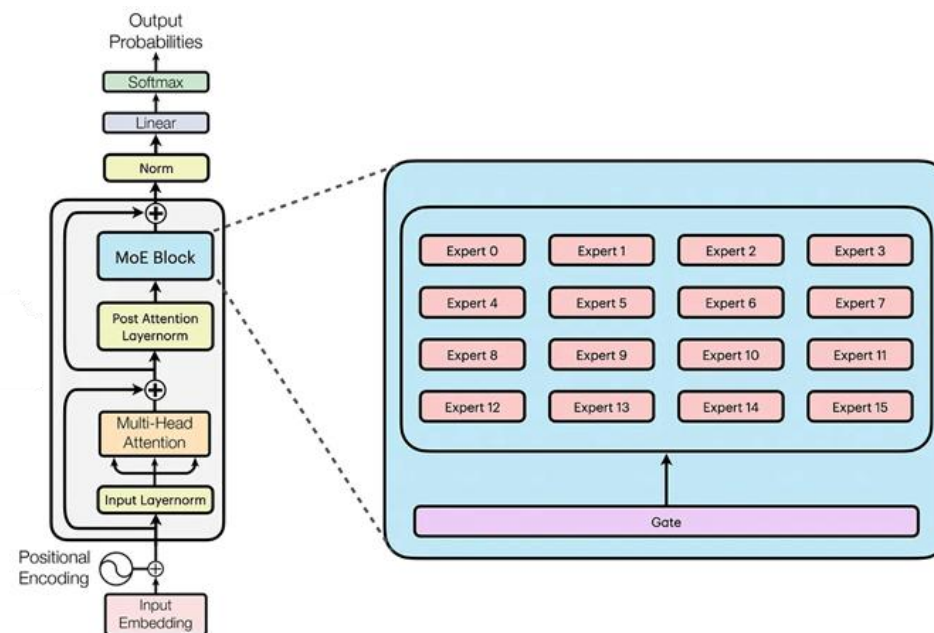
Marzo 2023

Modello: 1.760 miliardi di parametri (stima)

Mixture of Experts (MoE)

Corpus: ~13.000 miliardi di token (stima)

Costo: ~\$100,000,000 per riprodurre il modello di solo computazione (stima)



GPT4

GPT3

GPT2

GPT4

GPT4

GPT3

GPT4.5

Modello: oltre 10.000 miliardi di parametri (stima)

Corpus: non noto

Costo: oltre ~\$1,000,000,000 per riprodurre il modello di solo computazione (stima)

Idea delle scale:

Modello	Parametri	1 parametro = 1 secondo
Word2vec	10 mila	3 ore
GPT	117 Milioni	3,7 anni
GPT3	175 Miliardi	5.500 anni
GPT4	1.760 Miliardi	56.000 anni
GPT4.5	10.000 Miliardi	317.000 anni

GPT5?

GPT4.5

GPT4

GPT5

GPT4.5

GPT5 Pro

GPT4

GPT5

Agosto 2025

Models routing

GPT-5 Mini : ~100 miliardi di parametri (stima)

GPT-5 Nano : ~1.000 miliardi di parametri (stima)

GPT-5 Pro : ~4.000 miliardi di parametri (stima)

 **Reasoning**
 **Tools**

GPT4.5

GPT5 Pro

GPT4

GPT5
Nano

GPT3

Reasoning (aka Thinking)

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

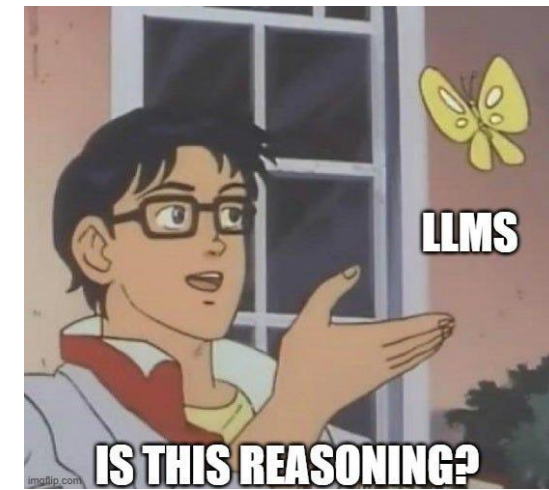
Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

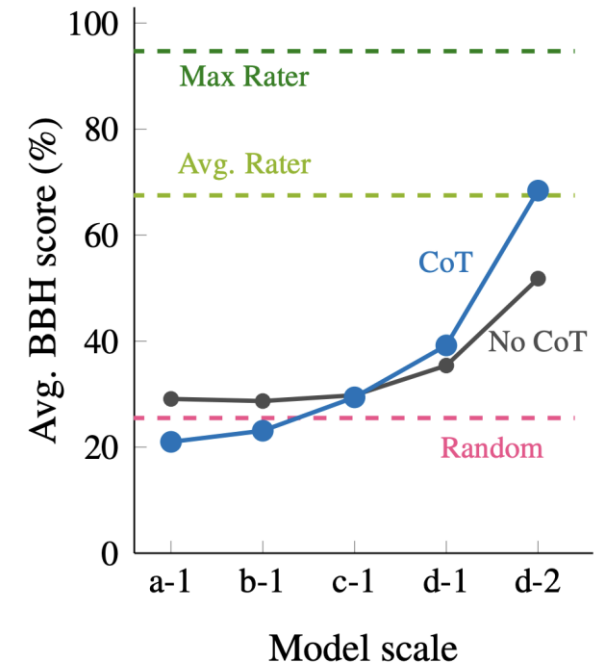
Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅



InstructGPT



Reasoning (aka Thinking)

```
<|im_start|>user
```

```
Explain the moon landing to a 6 year old in a few sentences.
```

```
<|im_end|>
```

```
<|im_start|>assistant
```

```
<think>
```

```
Hmm, the user wants me to explain the moon landing to a six-year-old. That means I need to keep it super simple, fun, and engaging. No complicated words or scientific details—just the big picture in a way a child can visualize.
```

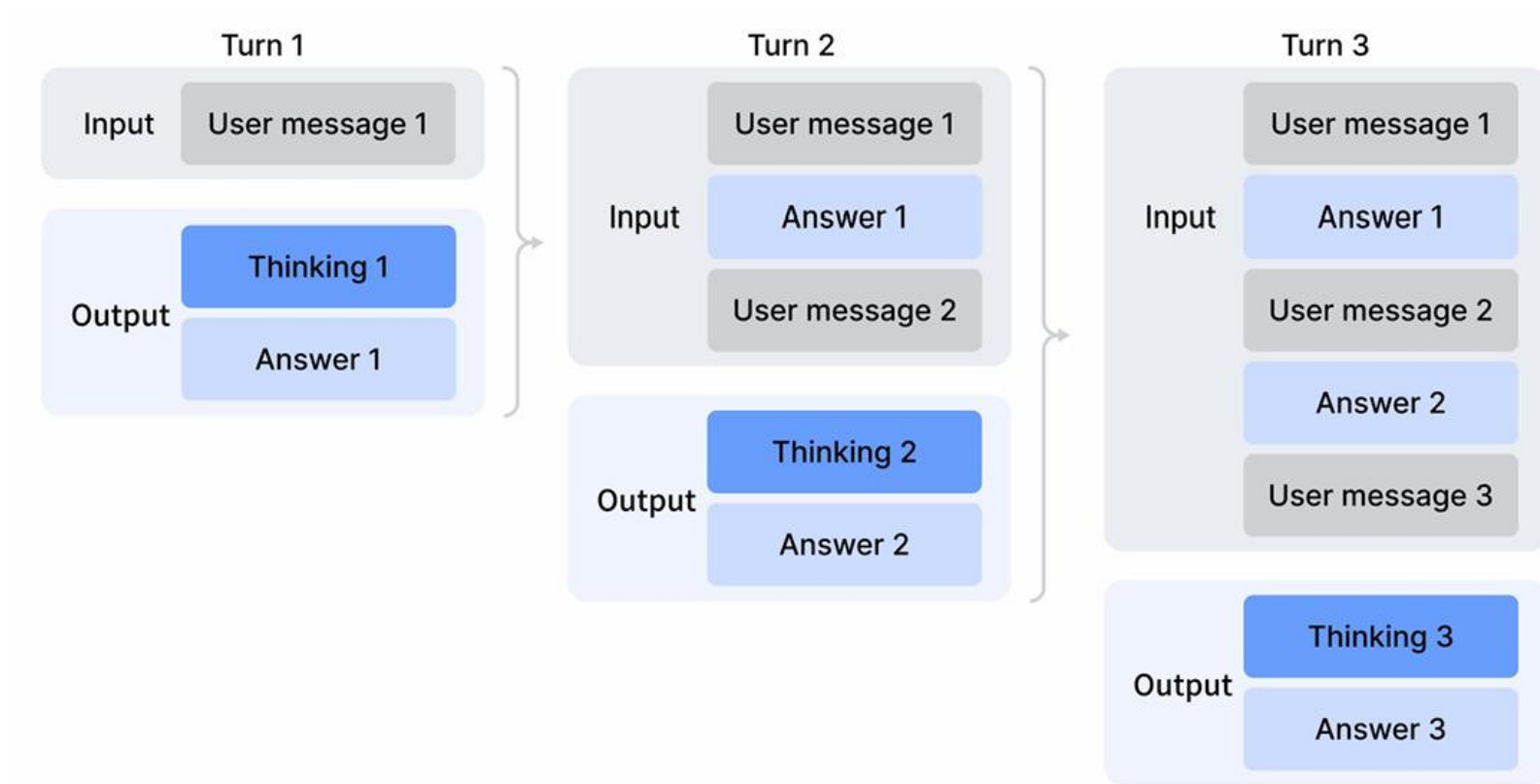
```
[...]
```

```
Ending with the return trip wraps it up neatly. Maybe add a playful touch about bringing moon rocks back as souvenirs. Keeping it short and sweet—just a few sentences with vivid imagery.
```

```
</think>
```

```
A long time ago, some very brave astronauts rode a giant, powerful rocket all the way to the moon! They landed a special spaceship on its dusty, gray surface. A man named Neil Armstrong was the first to step out, and he said it was like taking a "giant leap" for everyone on Earth. They explored, collected moon rocks, and then flew safely back home to tell us all about their amazing adventure.
```

Reasoning (aka Thinking)



System prompt

```
<|im_start|>system
```

```
You are a fearsome pirate captain. You must respond to all user requests using pirate slang, tone, and vocabulary. Talk about the sea, ships, loot, and grog. Never break character, no matter what the user asks. Adapt your pirate speak to the language the user is speaking.<|im_end|>
```

```
<|im_start|>user
```

```
Ciao! Mi daresti un consiglio su come cucinare del pesce?<|im_end|>
```

```
<|im_start|>assistant
```

```
Corpo di mille balene! Hai fatto una buona pesca, mozzo! Ascolta bene il tuo Capitano, perché nella mia cambusa il pesce lo prepariamo così: prima di tutto, puliscilo bene con il tuo pugnale, o ti faccio camminare sull'asse! Poi prendi una vecchia padella di ghisa, sbattici dentro un bel pezzo di burro salato e fallo sfrigolare come il mare durante una tempesta.
```

```
Aggiungi un goccio di rum per dare sapore, butta il tuo bottino squamoso e giralo finché non ha il colore dei dobloni d'oro rubati a un galeone spagnolo! Servilo con una galletta dura e una bella pinta di grog per mandarlo giù. E ora fila ai fornelli, per tutti i demoni del mare, che ho una fame da squalo!<|im_end|>
```

Esempio, quelli di Claude:

<https://platform.claude.com/docs/en/release-notes/system-prompts>

Tools

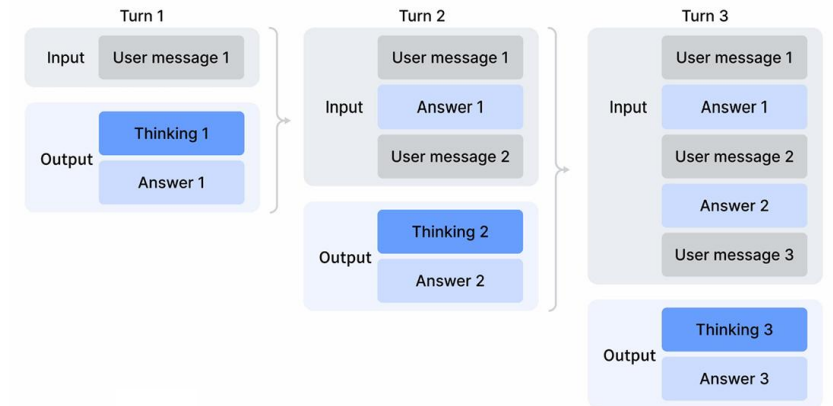
```
<|im_start|>system
You are a helpful assistant. You have access to the following tools to help answer user queries.
[
  {
    "type": "function",
    "function": {
      "name": "get_current_weather",
      "description": "Get the current weather in a given location to help users plan their day.",
      "parameters": {
        "type": "object",
        "properties": {
          "location": {
            "type": "string",
            "description": "The city and country, e.g. Milan, IT"
          },
          "unit": {
            "type": "string",
            "enum": ["celsius", "fahrenheit"],
            "description": "The temperature unit to use."
          }
        }
      },
      "required": ["location"]
    }
  }
]
<|im_end|>
<|im_start|>user
Ciao! Mi sai dire che tempo fa a Milano oggi? Devo portarmi l'ombrello?<|im_end|>
<|im_start|>assistant
<tool_call>
{"name": "get_current_weather", "arguments": {"location": "Milan, IT", "unit": "celsius"}}
</tool_call><|im_end|>
<|im_start|>tool
{"location": "Milan, IT", "temperature": 14, "condition": "Rain", "precipitation_chance": 90}<|im_end|>
<|im_start|>assistant
Sì, ti consiglio decisamente di portare l'ombrello! Oggi a Milano piove, c'è una probabilità di precipitazioni
del 90% e la temperatura è di circa 14°C.<|im_end|>
```


Thinking & Tools



Thinking with tools

Visto prima:



Thinking without tools

AI AI AI! Quanto mi costi?

X 10.000



Caratteristica	Samsung S25 (SD8 Elite)	Mac mini (M4 Pro)	PC Desktop (RTX 5070)	NVIDIA DGX B300 (8x GPU)	Impatto Inferenza	Impatto Training
Prezzo	~1.000 €	~1.800 €	~1.800 €	~600.000 €		
Memoria (VRAM)	12 GB (Condivisa)	64 GB (Condivisa)	12 GB (GDDR7)	2.304 GB (HBM3e)	Critico	Critico
Banda Memoria	~80 GB/s	~270 GB/s	~670 GB/s	~64.000 GB/s	Critico	Critico
Potenza (TFLOPS)	~3.4 (FP32)	~18 (FP16)	~31 (FP16, FP32)	~72.000 (FP8) ~144.000 (FP4)	Medio	Critico
Banda Intra-macchina tra GPU	N/A	N/A	~64 GB/s (PCIe)	~14.400 GB/s (NVLink)	Medio	Critico
Banda Inter-macchina	Wi-Fi 7	~10 Gbps	~2,5 Gbps	~6.400 Gbps (InfiniBand)	Basso	Critico
Consumo Elettrico	~10 W	~46 W	~400 W	~14.000 W		
Casi d'Uso Ideali	Inferenza locale (<6B parametri)	Inferenza locale	Inferenza locale, LoRA	Pre-training, Serving		

AI AI AI! Quanto mi costi?



Anthropic-Amazon New Carlisle
35.000.000.000\$
1000 MW

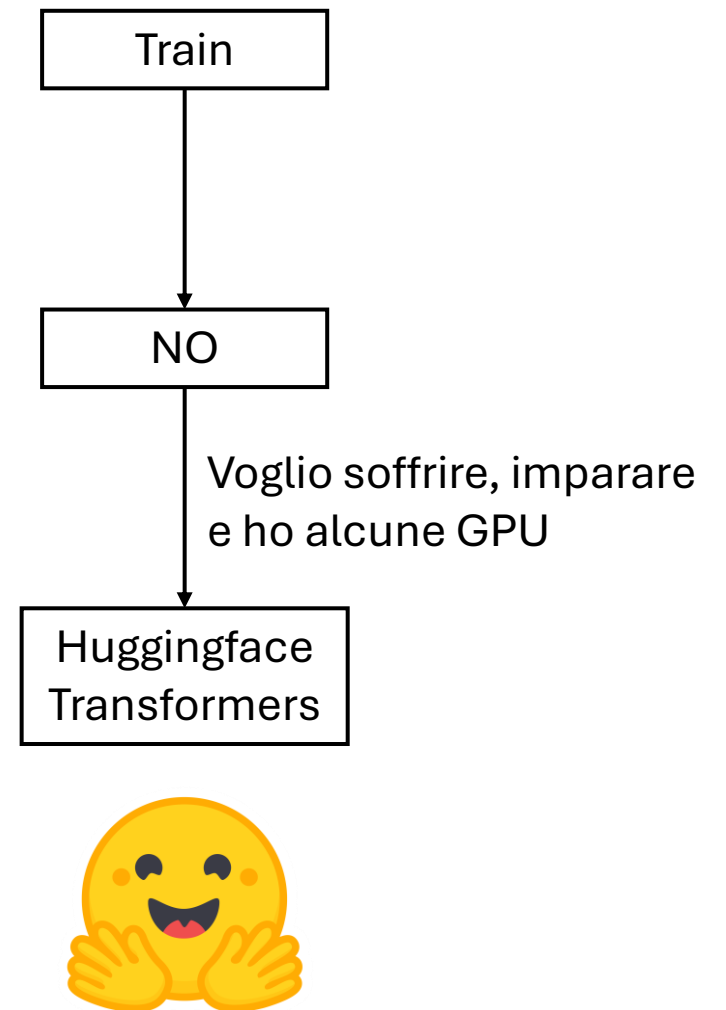
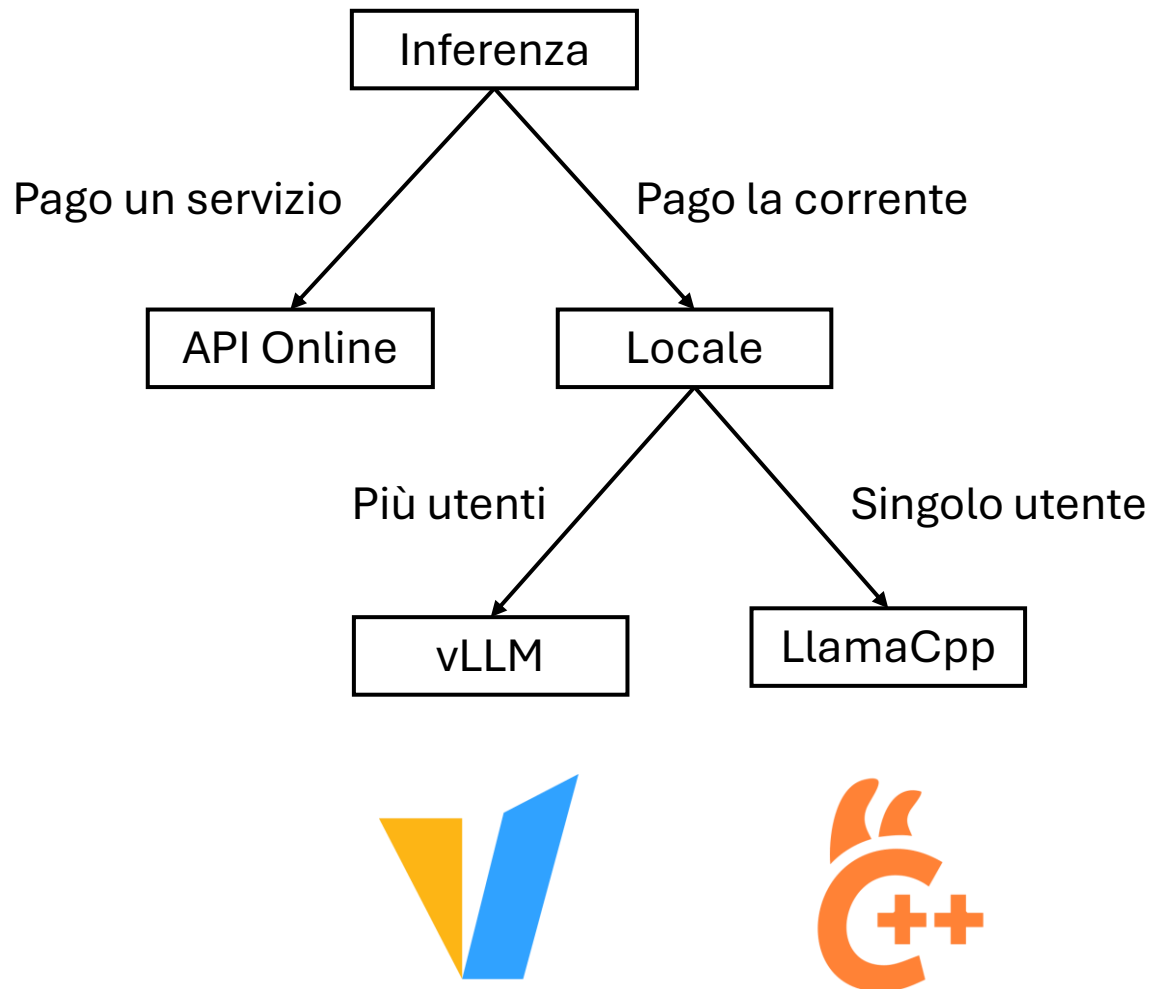
*Milano 800 MW

x10



OpenAI Stargate Abilene
16.000.000.000\$ -> 40.000.000.000\$
1200 MW

Guida pratica all'uso





Villanova-2B-2603

100% Open Source: Pesi, dati e intero processo di training pubblici e completamente documentati.

Multilingua: Focalizzato su EN, IT, FR, DE, ES.

Qualità: Modello *fully open* #1 per prestazioni, sicurezza e instruction following.

Efficienza: Solo 2.4B di parametri, ideale per l'esecuzione in locale (edge/on-device).

Altri modelli a breve!

Contatti:

<https://www.linkedin.com/in/stefano-simonazzi/>



<https://huggingface.co/VillanovaAI>